

Official Textbooks for Huawei ICT Academy



CLOUD COMPUTING TECHNOLOGY

Huawei Technologies Co., Ltd.

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Cloud Computing Technology

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Huawei Technologies Co., Ltd.
Hangzhou, Zhejiang, China



This work was supported by Huawei Technologies Co., Ltd.
ISBN 978-981-19-3025-6 ISBN 978-981-19-3026-3 (eBook)
<https://doi.org/10.1007/978-981-19-3026-3>

Jointly published with Posts & Telecom Press, Beijing, China
The print edition is not for sale in China (Mainland). Customers from China (Mainland) please order the print book from: Posts & Telecom Press.

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Preface

The sudden outbreak of novel coronavirus pneumonia (2019-nCoV) in 2020 has greatly affected people's lives. During the severe epidemic, everyone was quarantined at home and in the community, unable to go to work and school as usual. As a university teacher, the editor originally worried that normal teaching could not be carried out. Fortunately, with the help of the Internet and cloud services, even when the campus is closed, teaching and work in the university can still continue, teachers can teach online, and students can learn online. Cloud computing technology plays an important role in the information technology that has made a major contribution to the fight against the epidemic. As far as teaching is concerned, the communication between teachers and students is through the cloud: the teacher uploads the slides and teaching materials into the cloud, and the students obtain the materials from the cloud. The teaching cloud platform provides various functions such as sign-in, study, homework, and test. The communication among teachers and between teachers and students is through instant messaging software such as WeChat and QQ (the backend of these software is the cloud), and the meeting is through cloud conference systems such as Zoom, Tencent Meeting, and Webex. With the help of cloud computing technology, we can still teach and work normally during the epidemic, and this book can be completed.

Cloud computing technology has changed people's lifestyles, helped improve people's convenience and quality of life, and made a better future. Its emergence is a realistic portrayal of "technology changes life and changes the future." However, cloud computing technology itself integrates a variety of information technologies and is relatively complex. It is still difficult for beginners to have a more complete understanding of cloud computing technology. With regard to this reason, East China Normal University cooperated with Huawei to write this book on the basis of Huawei's certified cloud computing series of training materials.

There are eight chapters in this book. Chapter 1 introduces an overview of cloud computing. It includes common cloud computing scenarios in life. The characteristics, definition, origin and development of cloud computing, as well as advantages and classification of cloud computing are discussed in this chapter. It also mentioned

various types of supporting technologies for cloud computing and three perspectives on comprehending the business model, computing model and implementation of cloud computing. It also presents the state-of-art open source approach in cloud computing. Chapter 2 introduces the cloud computing system, starting from the four aspects of cloud infrastructure mechanism, cloud management mechanism, cloud security mechanism, and basic cloud architecture, and discusses some of the main technical mechanisms for building cloud technology architecture. Chapter 3 introduces virtualization technology, focusing on server virtualization technology, including its basic knowledge and supporting technologies, and discusses some actual virtualization products or applications based on theoretical knowledge, such as open-source virtual machine software KVM, Huawei FusionCompute cloud operating system, and desktop cloud. Chapter 4 introduces the basic knowledge of network in cloud computing, including an overview of computer networks, basic principles of computer networks, network interconnection equipment, network virtualization and software-defined networks, etc., so that readers can better understand some important concepts of computer networks, principles, equipment, and new network technologies supporting cloud computing. Chapter 5 introduces the basic knowledge of storage in cloud computing and also introduces storage in cloud computing in detail from the basic knowledge of storage, basic storage units, network storage, storage reliability technology, storage virtualization, distributed storage, etc. Chapter 6 starts from practice, introduces the popular open-source cloud operating system framework OpenStack, analyzes various components of OpenStack, discusses its operating mechanism, and enables readers to master the skills of implementing and managing OpenStack. Chapter 7 introduces the container technology that is widely concerned in cloud computing. It introduces the knowledge of container technology and container orchestration from the aspects of platform architecture, basic core functions, network, security, and resource management in a theoretical as well as practical model, which is convenient for readers to have a comprehensive understanding of Docker and Kubernetes ecosystem. Chapter 8 introduces the development status of cloud computing at home and abroad, analyzes and predicts its development trend, and discusses and analyzes the relationship between the Internet of Things, big data, artificial intelligence, 5G and other popular cutting-edge technologies, and cloud computing. It also introduces additional emerging technologies, such as edge computing and fog computing, microservices, and serverless computing.

This book was edited by Huawei Technologies Co., Ltd. The specific authors are as follows: Wang Wei formulated the outline and was responsible for the overall draft of the book, Zheng Kai was responsible for the main review, and Yang Lei provided training handouts. Chapters 1, 2, 4, and 8 are written by Zheng Kai. Chapters 3 and 5 are written by Huang Libo, and Chaps. 6 and 7 are written by Xu Yanjun.

In the process of writing this book, the editors have referred to and cited many works or papers by colleagues at home and abroad (see references for details), and I would like to express my gratitude to them. At the same time, I sincerely thank the training teachers and engineers of Huawei for their help and support.

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About the Author

Huawei Technologies Co., Ltd. Founded in 1987, Huawei is a leading global provider of information and communications technology (ICT) infrastructure and smart devices. We have approximately 197,000 employees and we operate in over 170 countries and regions, serving more than three billion people around the world.

Huawei's mission is to bring digital to every person, home and organization for a fully connected, intelligent world. To this end, we will: drive ubiquitous connectivity and promote equal access to networks to lay the foundation for the intelligent world; provide the ultimate computing power to deliver ubiquitous cloud and intelligence; build powerful digital platforms to help all industries and organizations become more agile, efficient, and dynamic; redefine user experience with AI, offering consumers more personalized and intelligent experiences across all scenarios, including home, travel, office, entertainment, and fitness & health.

Chapter 1

Introduction to Cloud Computing

Computing



This chapter is an overview of cloud computing, including common cloud computing scenarios in life, the characteristics, definitions, origins and development of cloud computing, the advantages and classification of cloud computing, various supporting technologies of cloud computing. The business model, computing model, and implementation of cloud computing are three perspectives for understanding, as well as open source methods that are currently very popular in the cloud computing field. Through the study of this chapter, I hope that readers have a clearer understanding of the general picture of cloud computing and lay the foundation for the in-depth study of the following chapters.

1.1 Ubiquitous Cloud Computing

As a representative of a new technology, cloud computing, like internet, has closely penetrated into our daily life. For example, we want to share an electronic material of hundreds of Mb with a friend from distance place, what happens if it exceeds the limitation of email attachment size? In the past, we generally used express delivery of storage media such as CDs, flash drives, or mobile hard drives which is time consuming and cost more work. Now we have a much more convenient way with the help of cloud storage service such as Baidu disk. Just put the data file into your own cloud disk and send the sharing link and access password to the recipient. The recipient can obtain the shared data file anytime and anywhere via the Internet. Another example is that an organizer wants to hold a special meeting while the participants are located all over the country. In an epidemic prevention and control situation, having participants gather by transport from all over the country for an on-site meeting not only takes considerable time and expense to travel back and forth, but also increases the risk of spreading the epidemic. Therefore, people will prioritize ZOOM meeting, Tencent meeting, or Webex as an option to hold online meeting. Participants only need to use the Internet to perform simple operations using a

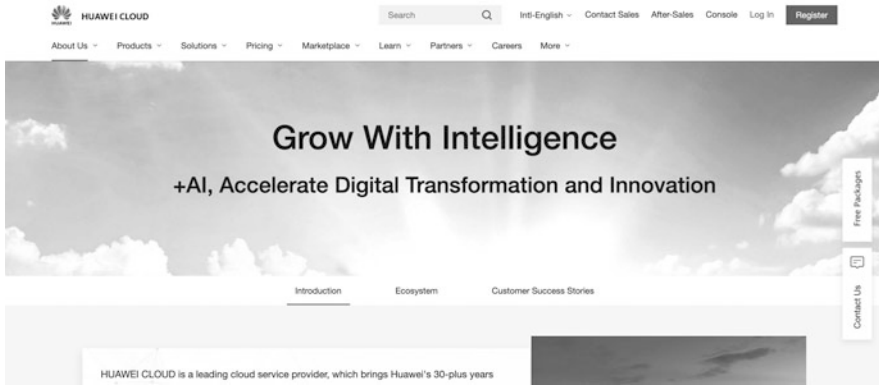


Fig. 1.1 Huawei cloud website

browser, and they can quickly and efficiently share voice, data files, and videos with participants in different geographical locations. In fact, participants in a cloud conference only need to have a device (computer, mobile phone, tablet, etc.) that can access the Internet that can be used normally to achieve online communication and video conferences without having to care about the complex technologies such as data transmission and data processing, all of which are provided by cloud conference service providers.

Such a way of preparing resources in advance and using these resources to perform specific tasks through specific technologies anytime and anywhere is generally a cloud computing type. The provider is a cloud service provider such as Huawei's public cloud. Let's take a look at the Huawei cloud website as shown in Fig. 1.1.

Under "Product" → "Fundamental services", we can find computing, storage, network, database, container services, etc. These divisions can be divided into different subdivision types. Take a popular service—ECS, an elastic cloud server, as an example, as shown in Figs. 1.2 and 1.3.

The elastic cloud server on the website is actually a virtual server (we will introduce it later). Similar to our own purchase of computer, the website provides different grades and types of cloud server instances to choose from. The configuration includes parameters such as the number of CPU, frequency, memory, and network bandwidth. Users can choose the most effective cloud server according to their needs. In fact, buying a cloud server instance is like buying a physical machine. You can complete most of the work that can be done on a physical machine, such as editing documents, sending emails, or working together. It's just that the cloud server is not in front of you, but on the far end of the network (cloud). In addition, the cloud server also has some advantages that the local physical machine does not have. For example, the access to the cloud server is not restricted by time and place. As long as there is Internet, it can be used anytime and anywhere. And the equipment (terminals) for operating the cloud server can be varied. For example, the user can

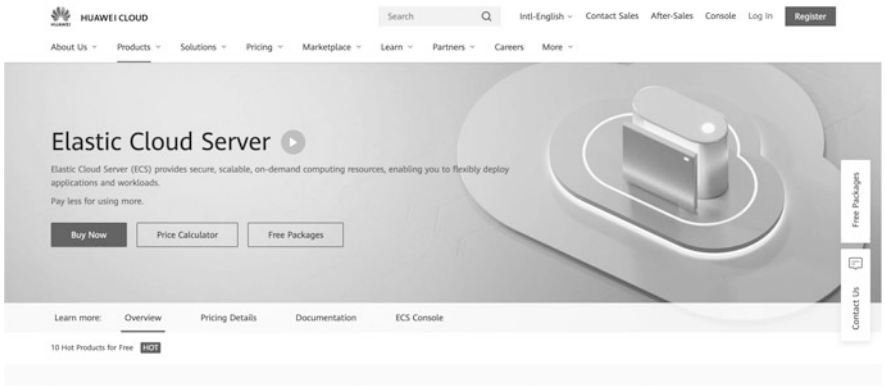


Fig. 1.2 Elastic cloud server

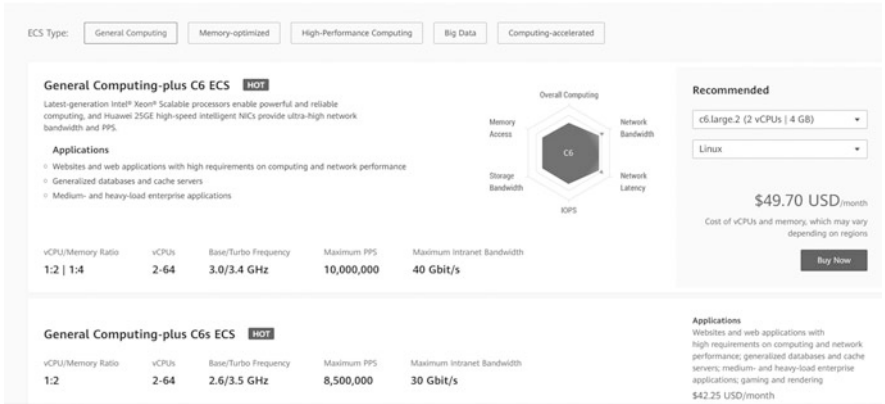


Fig. 1.3 Elastic cloud server specifications

operate the cloud server through a personal computer (PC), mobile phone, etc., and can modify or expand the performance configuration of the cloud server if necessary.

In addition to providing cloud servers to users, cloud service providers generally provide some other cloud services. For example, on Huawei’s public cloud, users who need to build a website can purchase the cloud speed website building service, which can help users quickly complete the construction of the website; users who need to store data can purchase object storage services or cloud hard drives. More advanced services also include artificial intelligence (Artificial Intelligence, AI) functions such as face recognition, voice recognition, image recognition, or text recognition.

In short, cloud computing allows us to use IT services like water and electricity as soon as the user turns on the faucet, water rushes out. This is because the water plant has sent water into the pipeline network (water network) that connects thousands of households; electricity is similar. For cloud computing, cloud service providers have

prepared all resources and services for users, and users can use them via the Internet. The Internet here is similar to the previous water network, and the tap can be a browser or a mobile application (App).

In fact, cloud services around you can be seen everywhere. In addition to the examples introduced in the previous article, other cloud services such as automatic backup of mobile phones, Youdao Cloud Notes and NetEase Cloud Music are all cloud services around us. At present, mainstream mobile phone manufacturers such as Huawei and Apple provide cloud-based mobile phone backup and recovery services. Users can back up files on the mobile phone to a data center in the cloud. After replacing the phone, you can restore your data to the new phone using your account and password. Youdao Cloud Notes is a product launched by NetEase, which provides online document creation and editing functions. When the user needs to record his inspiration at a certain moment, but unfortunately finds that there is no paper and pen around, the user can use Youdao Cloud Notes to record the inspiration online. Another advantage of this product is that no matter when and where, no matter what terminal the user uses (personal computer, mobile phone, etc.), online data can be edited, shared, and collaborated anytime, anywhere, and every edit can be done immediately Sync to the cloud. Music lovers may like NetEase Cloud Music App, through which songs can be listened to online and played at any time.

With the rapid development of cloud computing technology, similar cloud services will increasingly penetrate our daily lives. The spring weather turns rain, moisturizing things silently. We can truly feel the convenience of cloud computing technology in our lives.

1.2 The Properties of Cloud Computing

Cloud computing, as a new computing model, mainly has the following characteristics.

1.2.1 *On-Demand Self-Service*

Speaking of on-demand self-service, the first thing comes to your mind is supermarket. Every customer can collect goods according to their requirements. If it is the same type of goods, you can check description, price, and brand information to decide whether to buy or which one to buy. On-demand self-service is one of the mail characteristics of cloud computing. We will later introduce the Infrastructure as a Service (IaaS), platform as a service (PaaS) and Software as a Service (SaaS) model. Users can choose among one of these models based on their necessity. After selecting the mode, there will generally be different configurations to choose from, and users can purchase the services according to their needs. The entire process is

generally self-service and does not require third-party intervention unless you have a problem that requires consultation. As shown in Fig. 1.3, Huawei's elastic cloud server specifications for public clouds have many different configurations of cloud server instances to choose from.

On-demand self-service is premised on knowing your needs and which products will address them. This requires the relevant expertise of users using cloud computing. Users who do not have the knowledge and capabilities to use cloud services can consult a cloud service provider or turn to a relevant professional services provider.

1.2.2 Extensive Network Access

Another feature of cloud computing is that all clouds must rely on network connectivity. It can be said that the network is the foundation of cloud computing. Especially the Internet, the cloud is always inseparable from the Internet. The Internet provides remote, anytime, anywhere access to IT resources. Some people even think of cloud computing as "Internet plus computing," and network access is an intrinsic property of cloud computing.

Although most cloud access is over the Internet, cloud users also have the option of using a private channel to access the cloud. The level of service for network connectivity between cloud users and cloud service providers (quality of service, Quality of Service, QoS) depends on the Internet Service Provider (ISP) that provides them with network access.

In today's society, the Internet can cover almost every corner of the world, we can connect to the Internet through a variety of digital terminals, such as personal computers and mobile phones and connect to the cloud through the Internet, using cloud services. Therefore, extensive network access is an important feature of cloud computing. This can either be a wired network or a wireless network such as a Wi-Fi network. In short, without the network, there would be no cloud computing.

1.2.3 Resource Pooling

Resource pooling is one of the prerequisites for on-demand self-service, through resource pooling can not only put similar goods together, but also can refine the units of goods. Slightly large-scale supermarkets will generally be divided into fresh areas, fruit and vegetable areas, daily necessities areas and other areas to facilitate customers to quickly find their own needs of goods, but this form is not a pool of resources, can only be regarded as a classification of resources. So what is pooling resources? In addition to converting similar resources into resource pools, resource pooling requires the decomposition of all resources into smaller units. If we buy our own hard drives, a mechanical drive (Hard Disk, HDD) often has a few terabytes (TB, 1TB, 1012B); solid-state drives (Solid State Drive, SSDs) have a slightly

smaller capacity, and an SSD typically has a capacity of 128 to 512GB (Gigabytes, 1GB, 109B). Storage pooling cannot be measured in the number of hard drives because a hard drive has a large capacity, some applications only need a few gigabytes (GB), allocating the capacity of a hard disk is obviously a huge waste. Therefore, the way to use resource pooling need to break the number of physical hard disk unit “one” and combined all the capacity of the hard disk, gathered into a “pool.” Then allocation can be assigned in smaller units such as “GB” as a unit. Users can apply for as much as they need.

The computing resources include CPU and memory. If the CPU is pooled, the smallest unit of the CPU that the user sees can be a virtual core, and the CPU manufacturer no longer reflects the physical attributes of AMD or Intel.

Another function of resource pooling is to screen the differences between different resources. After the storage resources containing the mechanical hard drive and the SSD are pooled, if the user requests a certain amount of storage space, which corresponds to the mechanical hard drive or SSD, or both, he cannot tell the difference. In cloud computing, resources that can be pooled include computing, storage, and networking. Computing resources include CPU and memory. If CPU is pooled, the smallest unit of the CPU that the user sees can be a virtual core, and no longer reflect physical attributes such as the CPU’s manufacturer being AMD or Intel.

1.2.4 Fast and Elastic Scaling

Fast elastic scaling is one of the characteristics of cloud computing and is often cited as one of the core reasons for attracting users to “embrace” cloud computing. Cloud users can automatically and transparently scale their IT resources according to their needs. For example, in order to deal with the sudden high traffic of hot events, users can temporary self-purchase a large number of virtual resources to expand capacity. When hotspot events “cool down” and access traffic tends decline, users can release these newly added virtual resources, which is typical of fast elastic scaling. Cloud providers with large IT resources can provide a wide range of elastic scaling.

Fast elastic scaling includes several types, and in addition to manual capacity expansion or reduction, cloud computing supports automatic scaling or reduction based on preset policies. Scaling can be an increase or decrease in the number of servers, or an increase or decrease in resources for a single server.

In cloud computing, the biggest benefit of fast elastic scaling for users is cost savings while keeping the business or application running smoothly. Enterprises can purchase small amounts of resources when they are in low initial demand, gradually increase their investment in resources as the size of the enterprise expands, or concentrate all resources on priority business use during special periods, and, if resources are not sufficient, immediately apply for additional resources and, after a special period, release new resources. Either scenario is convenient for the user.

1.2.5 Measurable Services

Measuring is not billing although measuring is the basis of billing. Among the services provided by cloud computing, most services need to be paid for, but there are also services that are free. For example, elastic scaling can be opened as a free service for users.

Metrology is the use of technology and other means to achieve unity and accurate and reliable measurement. It can be said that the services in cloud computing are all measurable, some are based on time, some are based on resource quotas, and some are based on traffic. Measuring service can help users to automatically control and optimize resource allocation accurately according to their own business. In cloud computing systems, there is generally a billing management system that is specifically used to collect and process usage data. It involves the settlement of cloud service providers and the billing of cloud users. The billing management system allows for the development of different pricing rules and can also customize the pricing model for each cloud user or each IT resource.

Billing can choose between prepaid use or pay after use. The latter payment type is divided into predefined limits and unlimited use. If the limit is set, they often appear in the form of quota. When the quota is exceeded, the billing management system can reject the cloud user's further use request. Assuming that a user's memory quota is 500GB, once the user's storage capacity in the cloud computing system reaches 500GB, new storage requests will be rejected.

Users can purchase services according to their needs and can clearly see the usage of their purchased services. For contract users, the type of product used, service quality requirements, cost per unit time, or cost per service request are usually specified in the contract.

Figure 1.4 shows the pricing standards of Huawei Elastic Cloud Server instances, which shows the pricing standards of virtual server instances with different configurations. In this example, they are charged monthly.

1.3 Definition of Cloud Computing

There are several definitions of cloud computing. There are many definitions of what cloud computing is.

Wikipedia: Cloud computing is an Internet-based computing method. In this way, shared hardware and software resources and information can be provided to computers and other devices on demand, just like water and electricity for everyday use, paid for on demand, without caring about their source. National Institute of Standards and Technology, NIST: Cloud computing is a pay-per-use model that provides usable, convenient, on-demand network access to configurable computing resource sharing pools (resources including storage, software, services) that can be delivered quickly with minimal administrative effort or little interaction with service providers.

General Computing-plus C6 ECS HOT					Applications Websites and web applications with high requirements on computing and network performance; generalized databases and cache servers; medium- and heavy-load enterprise applications; gaming and rendering
vCPU/Memory Ratio	vCPUs	Base/Turbo Frequency	Maximum PPS	Maximum Intranet Bandwidth	
1:2 1:4	2-64	3.0/3.4 GHz	10,000,000	40 Gbit/s	
General Computing S6 ECS HOT					Applications Websites and web applications with high requirements on PPS; small-scale databases and cache servers; light- and medium-load enterprise applications \$7.70 USD/month
vCPU/Memory Ratio	vCPUs	Base/Turbo Frequency	Maximum PPS	Maximum Intranet Bandwidth	
1:1 1:2 1:4	1-8	2.6/3.5 GHz	500,000	3 Gbit/s	
General Computing S3 ECS					Applications Websites and web applications; small-scale databases and cache servers; light- and medium-load enterprise applications \$7.70 USD/month
vCPU/Memory Ratio	vCPUs	Base/Turbo Frequency	Maximum PPS	Maximum Intranet Bandwidth	
1:1 1:2 1:4	1-16	2.2/3.0 GHz	300,000	4 Gbit/s	
General Computing-basic T6 ECS					Applications Microservices, low-latency interactive applications; small- and medium-scale databases; virtual desktops; generalized-load websites and web applications, including development, build, and stage environments, code repositories, and product prototypes \$6.13 USD/month
vCPU/Memory Ratio	vCPUs	Base/Turbo Frequency	Maximum PPS	Maximum Intranet Bandwidth	
1:1 1:2 1:4	1-16	2.2/3.0 GHz	600,000	3 Gbit/s	

Fig. 1.4 Pricing standards for Huawei Elastic Cloud Server Instances

In the past, engineers used to use clouds to abstractly describe telecommunications networks or the Internet and underlying infrastructure when drawing pictures. The name of cloud computing has an inextricable origin. The “cloud” in cloud computing can be seen as a vast pool of IT resources where users can purchase the services they need on demand and pay for what they use.

Cloud computing is a broad concept, not a specific technology or standard, different people from different perspectives will have different understanding, there is no authoritative definition.

1. The definition of cloud computing by analysts

Early Merrill Lynch argued that cloud computing was the use of the Internet to run personal applications (E-mail, document processing, and presentations) and commercial applications (sales management, customer service, and financial management) on centrally managed servers. By sharing resources from these servers, such as storage and processing power, resources can be used more efficiently and costs can be reduced by 80% to 90%. Information Week, on the other hand, defines cloud computing more broadly: cloud computing is an environment in which any IT resource can be delivered as a service. The media is also interested in cloud computing. The Wall Street Journal, America’s best-selling magazine, is also keeping a close eye on the evolution of cloud computing. It argues that cloud computing enables enterprises to gain computing power, storage space, software applications, and data from very large data centers over the Internet. Customers pay only for the resources they use when necessary, avoiding the huge costs of building their own data centers and purchasing servers and storage devices.

2. The definition of cloud computing by enterprises

IBM believes that cloud computing is a computing style based on the delivery of services, software, and processing power over public or private networks. Cloud computing focuses on the user experience, with the core separating the

delivery of computing services from the underlying technology. Cloud computing is also a way to share infrastructure, using pools of resources to connect public or private networks together to provide IT services to users. Eric Schmidt, Google's former CEO, argues that cloud computing distributes computing and data across a large number of distributed computers, making computing and storage capabilities highly scalable and allowing users to easily access applications and services over the network through a variety of access methods, such as computers and mobile phones. Its important feature is open, there will not be an enterprise can control and monopolize it. According to Kaifu Li, a former global vice president at Google, the entire Internet is a beautiful cloud where Internet users need to easily connect to any device, access any information, create content freely, and share it with friends. Cloud computing is based on open standards and services, the Internet as the center, to provide secure, fast, and convenient data storage and network computing services, so that the Internet "cloud" is to become every Netizen's data center and computing center. Cloud computing is actually Google's business model, and Google has been working hard to promote the concept.

Microsoft's approach to cloud computing is much more contradictory than Google's. If future computing power and software are all concentrated in the cloud, then clients don't need a lot of processing power, and Windows loses most of its power. As a result, Microsoft's approach has always been "cloud+end." Microsoft believes that the future of computing model is not just cloud computing. The "end" here refers to the client, which means that cloud computing must have a client to work with. "From an economic point of view, bandwidth, storage, and computing are not going to be free, and consumers need to find a model that fits what they need, so there must be end-of-the-line computing. In terms of communication supply and demand, although bandwidth has increased, content is also growing simultaneously, such as video and images. Bandwidth limitations are always there. From a technical point of view, the end of the computing power is strong, in order to bring users more exciting applications" said Dr. Yaqin Zhang, a former senior global vice president at Microsoft. Microsoft's definition of cloud computing is no different, it just underlines the importance of the "end" in cloud computing. Today, with the rise of Azure Cloud, Microsoft has embraced cloud computing across the scale.

The overview of cloud computing across the business market is shown in Fig. 1.5.

3. The definition of cloud computing by academia

In academia, Ian Foster, the father of grid computing, argues that cloud computing is a model of large-scale distributed computing driven by the economics of scale. In this model, abstract, virtualized, dynamically scalable, and managed computing power, storage, platforms, and services converge into a pool of resources that are delivered to external users on demand over the Internet. He believes that several key points of cloud computing are: high scalability; can be encapsulated as an abstract entity and provide different levels of service for

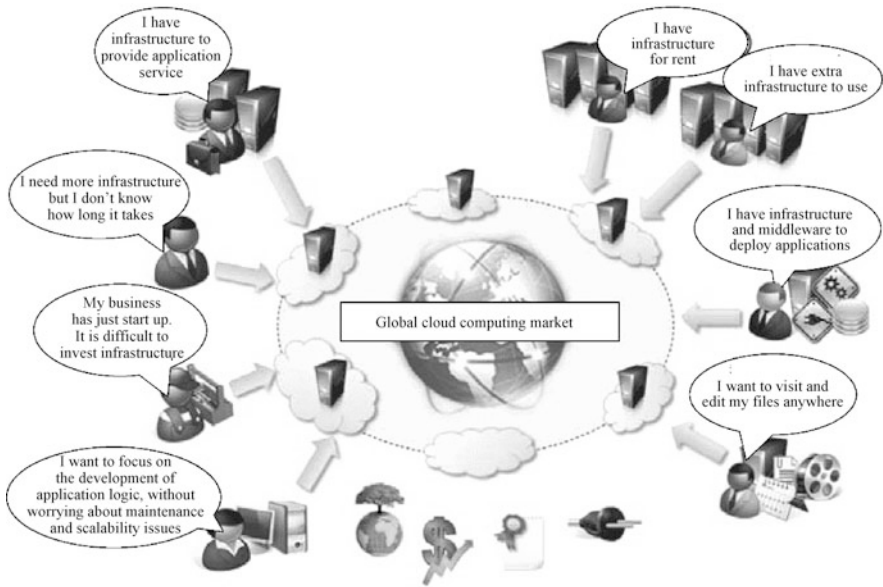


Fig. 1.5 Overview of cloud computing

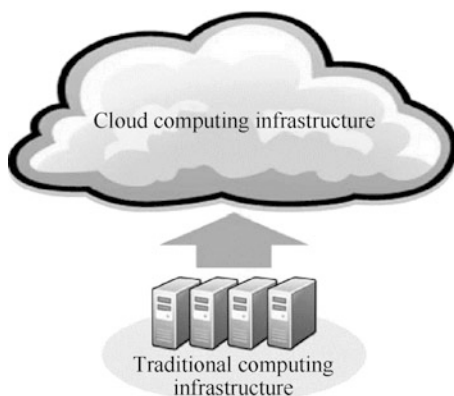
external users; economics resulting from scale; and services can be dynamically configured (via virtualization or other means) to deliver on demand.

Based on these different definitions, it's not hard to find out that the basic view of cloud computing is the same, but there are differences in the delimitation of certain areas. A more complete definition of cloud computing can be given from a comprehensive perspective: "Cloud computing is a computing model in which dynamically scalable and virtualized resources are delivered as services over the Internet." End-users don't need to know the details of the infrastructure in the cloud, do not need to have the appropriate expertise knowledge, do not need direct control, just pay attention to what resources they really need, and how to get the appropriate services over the network."

Zhu Jinzhi, who once worked at IBM, gave a relatively broad definition in his book *Smart Cloud Computing: The Platform of the Internet of Things* in order to cover cloud computing more comprehensively. The definition is as follows: "Cloud computing is a computing model: IT resources, data and applications are provided as services to users through the network." Its practical definition of "cloud" is a metaphorical method used to express the abstraction of complex infrastructure. Cloud computing is an abstraction of traditional computing infrastructure, so we choose to use "cloud" as a metaphor, as shown in Fig. 1.6.

Cloud computing starts with "software as a service," and then transforms all IT resources into services and provide to users. Think of cloud computing as a model that can easily access a common set of configurable computing resources (such as servers, storage devices, software, and services) through the network. These

Fig. 1.6 Cloud computing is an abstraction of traditional computing infrastructure



resources can be quickly provided and released, while minimizing management costs and the intervention of service providers.

We can look at cloud computing from two perspectives, the place where computing occurs and the form of resource supply. From the perspective of where computing occurs, cloud computing moves the operation of software from a personal computer (or desktop computer) to the cloud, that is, on a server or server cluster located in a “mysterious” geographic location. These servers or server clusters can be local, remote, or even far away. This seems to be a Client/Server (C/S) model, but cloud computing is not a traditional client/server model, but a huge improvement on this model. From the perspective of resource supply, cloud computing is a computing service, that is, all IT resources, including hardware, software, and architecture, are sold and charged as a service. For cloud computing, there are three main types of services: infrastructure as a service, providing hardware resources, similar to the traditional CPU, memory and I/O; platform as a service, providing an environment for software operation, similar to traditional operating system and programming framework in programming mode; software as a service, providing application software functions, similar to application software in traditional mode. In the cloud computing model, users no longer purchase or buy out certain hardware, system software, or application software to become the owner of these resources, but purchase the usage time of the resource, and consume according to the billing model such as paying for the length of use.

It can be seen that cloud computing treats all resources as services and consumes them in a pay-as-you-go manner, which is the characteristic of the host era. In the host era, all users are connected to the host through a display terminal and a network cable, and billing is based on the consumed CPU time and storage capacity. The difference is that in the host mode, the calculation occurs on one host; in the cloud computing mode, the calculation occurs in a server cluster or data center.

Therefore, cloud computing is both a new computing model and a new business model. It is a new computing model because all computing is organized

as a service; it is a new business model because the way users pay is different from the past, and pay according to what you use, which greatly reduces resource users' operating costs. It is not difficult to see that these two aspects of cloud computing rely on each other and are indispensable. Because only using resources as services can support the pay-as-you-go payment model; because the billing is based on what you use as you pay, resources can only be provided as services (not as packaged software or hardware). In fact, it can be said that cloud computing is a computing model, where computing boundary here is not determined by technical limitations, but by economic factors.

In a nutshell, cloud computing is the result of the hybrid evolution and integration of various concepts such as virtualization, utility computing, service computing, grid computing, and automatic computing. It started from mainframe computing and went through minicomputer computing, client/server computing, distributed computing, grid computing, and utility computing. It is not only a technological breakthrough (technical integration), but also a leap in business model (pay as much as you use, no waste). For users, cloud computing shields all the details of IT. Users do not need to have any knowledge or any control over the technical infrastructure of the services provided by the cloud, or even the system configuration and geographic location of the services provided. They only need to "turn on the switch" (Connect to the Internet) to enjoy the service.

It can be seen that cloud computing describes a new mode of supplying, consuming, and delivering IT services. This model is based on the Internet protocol and will inevitably involve the configuration of dynamically scalable and often virtualized resources. To some extent, cloud computing is a by-product of people's pursuit of easy access to remote computing resources.

The huge advantages of cloud computing in both technology and business model determine that it will become the leading technology and operating model of the IT industry in the future.

1.4 The Emergence and Development of Cloud Computing

The origin of cloud computing can be traced back to the concept of utility computing proposed by John McCarthy in 1961. He mentioned: "If the kind of computer I am advocating becomes the computer of the future, then the computer may one day be organized into a public facility like a telephone system...Computer facilities may become the basis of a new and important industry." Contains the initial thoughts of cloud computing. In the late 1990s, Salesforce took the lead in providing remote customized services to enterprises. In 2002, Amazon started operating the Amazon Web Services (AWS) platform, providing enterprises with services such as remotely customized storage, computing resources, and business functions. In 2006, Eric Schmidt, the CEO of Google, proposed the concept of "cloud computing."

In fact, the emergence of cloud computing is not isolated, but the product of the development of computer technology and communication technology to a certain

stage. Cloud computing technology is the product of a collection of various technologies.

There is a view that cloud computing is equivalent to the “Internet + computing” model, and the history of cloud computing is the history of the development of the Internet and computing models. So let’s briefly review the development history of the two.

1.4.1 The History of the Network and the Internet

In the early days, computers all operated on a stand-alone computer, and the calculation and transmission of data were all done on the local computer. The birth of the Internet connected these computers and ultimately connected the entire world. The following are some very representative milestones in the history of Internet development.

In 1969, ARPANET was born, and it is considered the predecessor of the Internet. There were only four nodes that first joined ARPANET, namely University of California, Los Angeles (UCLA); Stanford Research Institute (SRI); University of California, Santa Barbara (UC Santa Barbara); and University of Utah. The birth of ARPANET marked the beginning of the Internet era. In the following years, more and more nodes joined ARPANET, and more and more users in the non-military field. In 1983, out of security considerations, ARPANET separated 45 of these nodes to form a special military network called MILNET. The remaining nodes were used for civilian purposes.

In 1981, the first complete specification of TCP/IP was established, and the Internet has since had unified communication rules. TCP/IP is actually a collection of protocols, which includes Transmission Control Protocol (TCP), Internet Protocol (IP), and some other protocols. The earliest protocol used on ARPANET is called Network Control Protocol (NCP), but with the growth of ARPANET, NCP cannot meet the needs of large networks, while TCP/IP seems to be tailored for large or even giant network services. Therefore, in 1983, ARPANET replaced NCP with TCP/IP.

In 1983, ARPANET, PRNET, and SATNET were the three original networks that used TCP/IP communication. The first three networks switched to TCP/IP at the same time, marking the beginning of rapid development of the Internet.

In 1984, the Domain Name System (DNS) was born. After TCP/IP was adopted, the development of the Internet became more rapid, and more and more computers joined the network, and each computer used the digital IP address of the TCP/IP standard to identify each other. The common version 4 IP address (IPv4), an address corresponds to four bytes, and each byte is represented by a decimal number from 0 to 255. A typical IP address is in the form of 202.120.80.1. This type of digital IP address is not suitable for people to remember. It is like using an ID number to call the person you meet. I believe that few people can remember the ID number of everyone around them. Therefore, a new mechanism that is easy for people to remember is needed to replace the IP address to identify computers on the Internet.

Thus, DNS came into being. DNS can realize the mutual conversion between digital IP addresses and domain names that are easier for people to remember, which is equivalent to using short, easy-to-remember names instead of ID numbers, which greatly reduces the difficulty of remembering. For example, the previous IP address is 202.120.80.1, and the associated domain name is www.ecnu.edu.cn, which represents the official website of East China Normal University. The domain name is hierarchical. For example, the official website of Huawei www.huawei.com, the com on the far right represents the enterprise or company, the Huawei in the middle represents the enterprise name, and the www on the far left represents the default website server name. For general corporate websites, there will also be a top-level domain name (cn is China) that represents the country (or region) on the far right. Through DNS, we can use the domain name to visit the corresponding homepage worldwide.

In 1986, the modern mail routing system MERS was developed.

In 1989, the first commercial network operator PSINet was established. Prior to the establishment of PSINet, most networks were funded by the government or the military for military, industrial, or scientific research. The establishment of PSINet represents that the Internet has entered the era of commercial operation.

In 1990, the first Internet search engine Archie appeared. In the early days of the Internet, although there was relatively little information on the Internet, there were already many valuable files (data), but these files were scattered on various File Transfer Protocol (FTP) servers, which made it difficult for users turn up. Therefore, a search engine or search website is needed for indexing and searching. So Archie was developed. Using Archie can easily find the location of the FTP server where the file is located by the file name, and then download it to the local with tools such as FTP.

In 1991, the World Wide Web (WWW) was invented by Tim Berners-Lee, a scientist at the European Center for Particle Research. It was a landmark technology in the history of the Internet. Spread and interconnected on the Internet. Hypermedia can be documents, voice or video, and it is a new way of expressing information. After the birth of the World Wide Web, some great and landmark Internet companies were born, and various network applications that really changed people's lives began to emerge.

In 1995, e-commerce companies such as Amazon and eBay were established. In the history of Internet development, many companies have appeared, such as Yahoo! and Google. Amazon is the first Internet company to truly implement cloud computing. Amazon's early products sold were books. In order to process product information and user data, Amazon established a huge data center. In the United States, there is a "Black Friday" similar to "Double Eleven." On this day, Amazon needs to process a large amount of information, and all equipment in the data center will be turned on. But after this day, a lot of equipment will be idle. In order not to cause waste, Amazon will rent out the excess equipment. So in 2006, Amazon launched its first cloud computing product-EC2 (Elastic Compute Cloud).

In the late 1990s, the Internet was surging and experienced an "explosive" development. Domestic Internet companies "BAT" (Alibaba, Tencent, Baidu)

were established during this period. The Internet has allowed people to see its magic, leading to rapid development, spawning a large number of “bubbles,” and finally around 2000, the Internet bubble burst. However, after experiencing the bubble burst, the Internet has rapidly developed, and 2004 is also known as the “first year of social networking.” With the rapid development of domestic Internet applications, from 2000 to 2020, a large number of domestic Internet companies have rapidly developed and become industry “giants.” In addition to the BAT mentioned above, there are also [JD.com](#), Ant Financial, ByteDance, Pinduoduo, Meituan, NetEase, Sina, Sohu, Didi, Suning, Xiaomi, etc. These Internet companies have penetrated deeply into people’s daily lives. And even changed people’s lifestyle to a certain extent.

1.4.2 The History of Computing Models

Cloud computing does not appear suddenly, but a result of the development and evolution of past technologies and computing models. It may not be the ultimate result of computing models, but a model suitable for current business needs and technical feasibility. The following describes the emergence of cloud computing by analyzing the development history of computing models. Figure 1.7 shows the development history of cloud computing from the perspective of computing models.

Cloud computing is a master of multiple computing technologies and computing models. Therefore, it has similarities with many existing computing models. For example, the software runs in the cloud, and the way customers access cloud services through mobile terminals or clients is somewhat similar to the client/server model; the automatic scaling of cloud resources is somewhat similar to automatic computing; cloud computing gathers resources for customers to use, and grid computing,

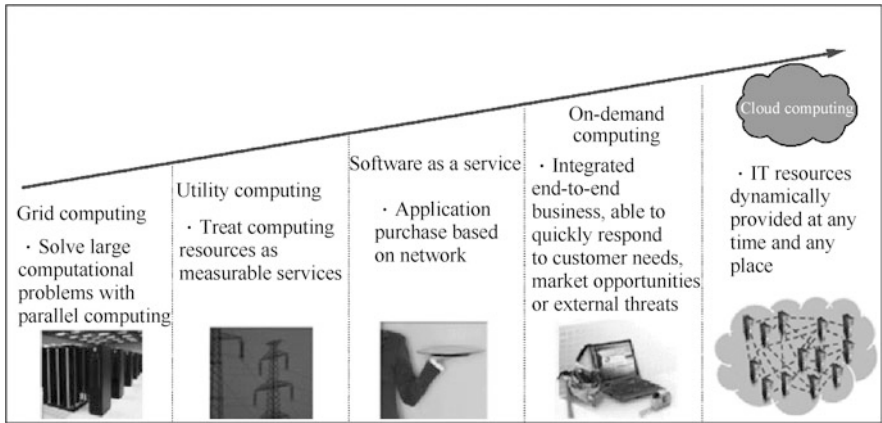


Fig. 1.7 Show the development history of cloud computing from the perspective of computing mode

which was once a smash hit, is somewhat similar; a large number of computing nodes in cloud computing work at the same time, which seems to be somewhat similar to parallel computing; nodes that make up the cloud are distributed in multiple locations and are somewhat similar to distributed computing. Billing based on usage is somewhat similar to utility computing.

Although cloud computing does have similarities with various computing models that we are familiar with, they are not exactly the same. In fact, there are huge differences between cloud computing and certain computing models.

1. Cloud computing and mainframe computing

As early as a few decades ago, shortly after the computer was invented, the computing model at that time had a “shadow” of cloud computing. In 1964, the world’s first mainframe System/360 was born, triggering a “revolution” in the computer and business fields.

The market for the mainframe system is mainly enterprise users, and these users generally have a variety of business systems that need to use the mainframe system. So IBM invented the virtualization technology, which divides a physical server into many different partitions, and each partition runs an operating system or a set of business systems. In this way, each enterprise only needs to deploy a mainframe system to meet the needs of all business systems. Since the system has experienced decades of development, its stability is high and it has the reputation of “never shut down.” The IBM mainframe is responsible for the most extensive and most important information and data processing tasks in the finance, communications, energy, and transportation industries. Before the emergence of cloud computing, more than 70% of the world’s corporate data was running on mainframe platforms, and most of the top companies in the world’s wealth rankings were using mainframes.

One characteristic of the mainframe is the concentration of resources, the concentration of computing and storage, which is a typical representative of the centralized computing model. Enterprises using mainframes will concentrate the business to be processed, usually in the form of batch processing and send them to the mainframe for processing. Most of the users of the mainframe use the terminal mode to connect to the mainframe, no data processing and storage are performed locally, and no measures such as patch management, firewall protection, and virus prevention are required. In fact, the mainframe system is the earliest “cloud, but these clouds are for specialized services, private networks, and specific fields.

Cloud computing and mainframe computing actually have many things in common, such as centralized management and billing based on usage. However, there is a big difference between cloud computing and mainframe computing. One of the important differences is the different user groups. The users of mainframe computing are usually large organizations and are prepared for key software, such as census, consumption statistics, enterprise resource planning (ERP), and financial transactions; while cloud computing is for the general public and can run a variety of large, medium, and small software. In addition,

the processing unit of mainframe computing is usually a single mainframe, while the processing unit of cloud computing is generally composed of a large number of IT resources in a cluster manner, and the processing capacity is much greater than that of the early single mainframe.

2. Utility calculation

Utility computing emerged with the development of the mainframe. Considering the high purchase cost of the host, some users can only rent it instead of buying it. So some people put forward the concept of utility computing, whose goal is to package the server and storage system for users to use, and to charge users according to the amount of resources actually used by the users. This model is similar to the provision of water, electricity, gas, and telephone services, enabling users to use computer resources as if plugging a light bulb into a lamp holder. This model eliminates the need for users to own resources in order to use services and can also achieve the goal by leasing resources. Utility computing can be regarded as the predecessor of cloud computing.

The actual application of utility computing is mainly represented by IBM. IBM leases its own host resources to different users according to time. The host is still stored in IBM's data center, and users use IBM's resources remotely or on site in IBM's data center. The key technology in utility calculation is resource usage measurement, which guarantees the accuracy of pay-per-use.

From the perspective of the billing model, cloud computing is exactly the same as utility computing. Utility computing packages IT resources into measurable services for users to use, that is, CPU, memory, network bandwidth, and storage capacity are all treated as traditional utility usage (such as telephone networks) for packaging. The biggest advantage of this computing model is that users do not need to pay in advance, nor do they need to buy out IT resources. For most of the small- and medium-sized enterprises, there is not enough capital and technology to construct an IT infrastructure comparable to the Fortune 500 companies. They welcome the concept of utility computing because utility computing enables them to access and use advanced information technology and resources like the Fortune 500 like companies.

Compared with cloud computing, utility computing only specifies the billing model of IT assets and does not limit other aspects of IT assets, such as technology, management, configuration, and security. There are many more factors to consider in cloud computing, and the billing model is only one of the factors.

3. Client/server model

From the perspective of service access mode, cloud computing does have the shadow of a client/server model: customers connect with the remote cloud through a certain device and use the services provided by the application software running in the cloud. However, behind this similarity, the "remote server" provided by cloud computing has unlimited computing power, unlimited storage capacity, and never crashes, and all software can run on it. Users can also publish their own software to the "remote server," and the "remote server" can automatically configure the required resources for the software and change it as

needed. In addition, cloud computing has its own set of models and rules (explained later), while the client/server model, on the other hand, refers to all Distributed Systems that can distinguish between a service provider (server) and a service requester (client).

4. Cluster computing

Since the cloud of cloud computing contains a large number of server clusters, it is very similar to cluster computing. However, cluster computing based on server clusters uses a tightly coupled group of computers to achieve a single purpose, while cloud computing can provide different supports according to the needs of users to achieve different purposes. In addition, cluster computing is limited distributed computing, which is not as complicated as the distributed computing faced by cloud computing. In addition, cluster computing does not consider interactive end-users, while cloud computing does. Obviously, cloud computing includes elements of server cluster computing.

5. Service computing

The service provided by cloud computing is called cloud service, which is naturally reminiscent of service computing. Service computing is also called service-oriented computing and has the same concept as SaaS described later. This computing model provides all applications as services, and users or other applications use these services instead of buying out or owning software. In the service computing mode, different services are relatively independent, loosely coupled, and freely combined. For service computing, discovery of service is the key point.

Cloud computing has largely adopted the technology and way of thinking of service computing, but there are still important differences between service computing and cloud computing. First, although service computing is generally implemented on the Internet, service computing does not necessarily have to be provided in the cloud. A single server, small-scale server clusters, and a limited range of network platforms can provide service computing. Secondly, service computing is generally limited to providing services at the software level, while cloud computing extends the concept of services to hardware and operating environments, including the concepts of IaaS and PaaS. In other words, the concept of cloud computing is more extensive than the concept of traditional service computing.

6. Personal computer and desktop computing

In the 1980s, with the development of computer technology, the volume and cost of computer hardware were greatly reduced, making it possible for individuals to own their personal computers. The appearance of personal computers has greatly promoted the development of the software industry, and various end-consumer-oriented software has emerged. Software running on personal computers requires a simple and easy-to-use operating system. The Windows operating system just meets the needs of the public, and it has occupied the market with the popularity of personal computers. Personal computers have their own independent storage space and processing capabilities. Although their performance is limited, they are sufficient for individual users within a period of

time. Personal computers can complete most of the personal computing needs; this model is also called desktop computing.

Before the advent of the Internet, the sales model of software and operating systems was an authorization model, that is, the software code was copied to a computer through a floppy disk or CD-ROM, and each copy required a payment to the software developer. After several years of development of this model, some problems appeared, such as high cost and cumbersome software upgrades. The purpose of the upgrade is to solve some of the previous problems or to use new features, but the upgrade process can sometimes be cumbersome. For a large enterprise, its IT department may need to manage hundreds of software, thousands of versions, and tens of thousands of computers. Each version of the software needs to be maintained, including problem tracking, patch management, version upgrades, and data backup. This is by no means a simple job.

7. Distributed computing

The personal computer did not solve the problem of data sharing and information exchange, so the network-Local Area Network (LAN) and later the Internet appeared. The network connects a large number of computers distributed in different geographical locations, including personal computers and servers (large mainframes and later medium and small mainframes). With so much computing power, can an application run on multiple computers to complete a computing task together? The answer is of course yes, this is distributed computing.

Distributed computing relies on distributed systems. A distributed system consists of multiple computers connected through a network. Each computer has its own processor and memory. These computers cooperate with each other to complete a goal or computing task together. Distributed computing is a large category, which includes many familiar computing modes and technologies, such as grid computing, P2P computing, client/server computing, and Browser/Server (B/S) computing. Of course, cloud computing is also included. In today's network age, there are very few non-distributed computing applications, and only some stand-alone applications fall into this category, such as word processing and stand-alone games.

8. Grid computing

One of the main functions of computers is to perform complex scientific calculations, and a "master" in this field is supercomputers, such as China's "Galaxy" series, "Dawn" series, "Tianhe" series, and "Shenwei-Light of Taihu Lake." In foreign countries, there are Japan's Fugaku, which ranks first in the global supercomputing rankings in 2020, and the US's Summit, which is the second. The computing model centered on supercomputers has obvious shortcomings: Although it is a "big Mac" with powerful processing capabilities, it is extremely expensive, and is usually only used by some state-level departments (such as aerospace, meteorology, and military industries) who have the ability to configure such equipment. As people increasingly need computers with more powerful data processing capabilities, people began to look for a low-cost

computing model with superior data processing capabilities. Finally, scientists found the answer, that is, grid computing.

Grid computing appeared in the 1990s. It is a new type of computing model that has been developed rapidly with the development of the Internet, specifically for complex scientific computing. This computing model uses the Internet to organize computers distributed in different geographical locations into a “virtual supercomputer.” Each computer participating in the calculation is a “node,” and the entire calculation is composed of thousands of “nodes,” so this calculation mode is called grid computing. In order to perform a calculation, grid computing first divides the data to be calculated into a number of “small pieces,” and then distributes these small pieces to each computer. Each computer executes its assigned task segment and returns the calculation result to the master control node of the calculation task after the task calculation is completed.

It can be said that grid computing is an extension of supercomputers and cluster computers. Its core is still trying to solve a huge single computing problem, which limits its application scenarios. In fact, in non-scientific fields, only a limited number of users need to use huge computing resources. Grid computing once became “hot” after entering the twenty-first century. Major IT companies have made many investments and attempts, but they have not found a suitable use scenario. In the end, grid computing has made a lot of progress in the academic field, including some standards and software platforms have been developed, but it has not been popularized in the commercial field.

To some extent, many things that grid computing must do are also things that cloud computing must do, but grid computing cannot be regarded as cloud computing. First of all, grid computing is mainly for scientific computing and simulation, while cloud computing is made for general public. Secondly, grid computing does not consider interactive end-users, while cloud computing must consider.

9. SaaS

SaaS is a computing model that uses software as a service. It is a mode of providing software through the Internet. Service providers uniformly deploy the software on their own servers. Users can order the required software services from the service provider through the Internet according to their actual needs and pay the service provider according to the number of services ordered and the length of time and obtain the services provided by the service provider through the Internet. Users no longer need to purchase software, but instead rent Web-based software from service providers to manage business activities, and there is no need to maintain the software. The service provider will have full authority to manage and maintain the software. While providing Internet software to users, service providers also provide offline operation of the software and local data storage, so that users can use the software and services they have ordered anytime, anywhere.

SaaS first appeared in 2000. At that time, with the vigorous development of the Internet, various new business models based on the Internet continued to emerge. For traditional software companies, SaaS is the most significant change.

This model turns the one-time software purchase income into continuous service income. Software providers no longer calculate how much software they sell but need to always pay attention to how many paying users there are. Therefore, software providers will pay close attention to their own service quality, and continuously improve their own service functions to enhance their own competitiveness. This model can reduce piracy and protect intellectual property rights, because all the code is at the service provider, users cannot obtain it, nor can it be cracked or decompiled.

In summary, in addition to utility computing, the computing models discussed above are all technical aspects, and the cloud computing model covers both technical and commercial aspects. This may be the biggest difference between cloud computing and the above-mentioned computing models.

10. The emergence of cloud computing

Looking back at the history of the development of computing models, it can be summarized as: Concentration-Decentralization-Concentration. In the early days, limited by technical conditions and cost factors, only a few companies could have computing power. Obviously, the computing model at that time could only be centralized. Later, with the miniaturization and low cost of computers, computing also became decentralized. Up to now, there is a trend toward centralization in computing, and this is cloud computing.

Users can use cloud computing to do many things. As we can see from the previous article, there are three basic services provided by cloud computing: the first one is hardware resource services; the second is operating environment services; and the third is software services. Then users can also use the cloud computing platform (also called cloud platform) in at least three ways: the first one is to use the cloud platform to save data (using the hardware resources provided by the cloud environment); the second is to run software on the cloud platform (using the cloud the operating environment of the environment); the third is to use software services on the cloud platform (using software services arranged on the cloud, such as maps, search, and mail).

The service provided by cloud computing is not only the IT resource itself, if it is nothing more than that, there is no need to develop cloud computing. Storage of data, running programs, and using software can be implemented on many platforms, and cloud computing is not required. The reason for using cloud computing is a result of the way and the ability to provide resources. Cloud computing has great advantages in resource provision methods and capabilities.

In addition to the scale advantages of the cloud platform mentioned above, another important advantage of cloud computing is elastic resource allocation. The resources provided by cloud services are more when need is high, and less when need is low. If we deploy an application software on the cloud, the cloud controller will dynamically adjust the resources allocated to the application according to the changes in customer needs of the software, so as to ensure that it can meet any sudden increase in customer demand at any time and at the same time avoid waste of resources when customer demand is low.

In addition, cloud platform also provides another advantage that most people may not know. Some operations can only have a powerful effect if they are placed in the cloud, while direct deployment on the business location or the user's client has limited or no effect. This is because technically, the desktop or server operation mode is no longer adequate for many challenges faced by IT systems, and these challenges can be solved in the cloud. For example, in terms of virus checking and killing, the antivirus software on the desktop has a lackluster checking and killing effect, and the highest antivirus efficiency can only reach 49% to 88% (according to the data of Arbor Networks in the United States). In addition, scanning and killing also occupies a large amount of computing resources of the personal computer, resulting in extremely low efficiency of the entire system. But moving the antivirus operation to the cloud can solve this problem. Note that moving antivirus operations to the cloud mentioned here is very different from the "cloud antivirus" promoted by some antivirus software companies. The cloud antivirus technology on the market refers to deploying antivirus software in the cloud and checking and killing remote clients through the network. The only advantage of this kind of cloud antivirus is that it is easier to update and maintain antivirus software, but the antivirus capability has not improved. However, if a variety of different antivirus software is deployed in the cloud and the network data is cross-checked and killed in the cloud, the antivirus efficiency can be increased to more than 96%, and it will not occupy the computing resources of the client.

Another example is, in the market, the operating modes of personal computers, servers, or clusters face difficulties in updating and maintaining. Only changing the functional role of a server (replacement of the system and software) requires a lot of effort and is prone to errors. The installation, configuration, upgrade, and other management operations of software distributed on various computers in the organization are a headache for many companies. Moving these services to the cloud can solve these problems.

In general, cloud computing mainly has the following four advantages:

- On demand, unlimited computing resources.
- Instantly available software/hardware resources and IT architecture.
- Charge by usage model.
- Processing environment that is difficult to provide by a single machine.

Although various service concepts with the title "cloud computing" emerge in an endless stream, not every service can be classified as a cloud computing service. How to judge whether a service is a true cloud computing service? Generally speaking, you should see whether the following three conditions are satisfied at the same time.

1. The service should be accessible anytime, anywhere. Users can use the services at anytime, anywhere, and through any device that can connect to the Internet, without having to consider the installation of applications or the implementation details of these services.

2. The service should always be online. Occasional problems may occur, but a true cloud computing service should always ensure its availability and reliability. That is, ensure that it can be accessed through the network at any time and provide services normally.
3. The service has a large enough user base. This is the so-called “multi-leasing,” a “leasing” in which a basic platform provides services to multiple users. Although there is no clear number to divide, it is only for a small number of users, that is, using cloud computing-related technologies to support its basic system architecture, it should not be classified as a cloud computing service. Because only a large user base will generate pressure to access massive amounts of data. This is the most fundamental reason for the emergence of cloud computing, and it is also one of the signs that cloud computing services are different from other Internet services.

1.4.3 The Driving Force of Cloud Computing

Cloud computing does not appear from nowhere. Its emergence is promoted by a variety of factors, which has a certain inevitability.

1. The improvement of internet broadband

Bandwidth is a necessary condition for the popularization of cloud computing. Since computing and storage are placed on the other side of the network, it is necessary to allow users to easily access these data. In recent years, with the popularization of the Internet, major network operators have also continued to improve their Internet infrastructure. On the one hand, the bandwidth of the core network is rapidly expanding; on the other hand, the network access of home and business users has also undergone essential changes. Take home users as an example. From the very beginning dial-up Internet access (speeds in the range of tens to hundreds of kilobits per second), to the later Asymmetric Digital Subscriber Line (ADSL) (speeds in the hundreds of thousands). Bits per second to several megabits per second), and then to the current fiber to the home (network speeds are tens to hundreds of megabits per second or even higher). The increase in bandwidth has changed the mode of use of the network and the types of network application. With the development of 4G/5G technology, network bandwidth will further increase until users do not perceive the limitation of bandwidth.

2. Technology maturity

There are many similarities between cloud computing and utility computing, but utility computing is not really popular because of the lack of sufficient operability. Any idea, if there is no practical way to realize it, will become a fantasy. The recognition of cloud computing by the public is also closely related to its technological maturity. Cloud computing corresponds to not one technology, but a combination of multiple technologies, which turns the concept of IT as a service into reality. At different levels, different technologies may be used.

These technologies are hidden in the background and are invisible to users. This is also the hidden part of the cloud. We can imagine a data center that provides cloud computing services as a huge factory filled with hundreds of servers and connected by intricate cables. Many intelligent applications are running on these servers. They can manage these servers efficiently, ensure that the system can automatically recover when server fails, and also ensure that the entire center is running at a very low cost.

3. The Development of Mobile Internet

The rapid development of the mobile Internet has led to a rapid increase in the number of digital mobile terminal devices represented by mobile phones and tablet computers. On average, everyone in the country has multiple digital mobile terminal devices that can access the Internet. How to manage the data in these devices has become a big problem. One is that these devices cannot all have strong computing power; the other is that data is scattered on each device, with duplication and redundancy, and the same data may also exist in both new and old versions. So the cloud computing model has become an ideal solution to this problem. For example, users may need to uniformly manage photos on computers, mobile phones, and digital cameras. Although they can be copied or transferred between devices, it is very troublesome. If these devices are connected to the Internet, the photos are synchronized to the cloud via the Internet, and the cloud-based photo management is performed, and the classification, update, synchronization, and access of the photos become very convenient.

4. The evolution of data center

For users, a data center is a “factory” that provides computing and storage capabilities on the other end of the Internet and is a “power plant” for the IT industry. Data centers are unfamiliar to ordinary Internet users, just as everyone who uses electricity does not care about how power plants operate. In fact, data centers are constantly evolving. Data centers can be divided into two types: one is to provide services to the Internet; the other is private to the enterprise and only open to the inside. Regardless of the type, the data center needs someone to operate it to ensure that it can provide services uninterrupted. According to a survey, more than 90% of 1000 organizations worldwide believe that they need to make major changes to their data centers in recent years. For them, the current challenges include expensive management costs, rapidly increasing energy consumption, rapidly increasing user demand, and inefficient use of IT resources. In view of these problems, data centers urgently need a new architecture and management concept, and cloud computing is a solution from the perspective of service providers.

5. Economic factor

When a product is technically feasible and has a wide range of needs, the only factor that determines its success or failure is the price, or user cost. The most fundamental factor in changing the computing mode is cost, and technology is the triggering condition. In the era of mainframes, the main reason for the use of centralized computing is that the cost of the mainframe is too high, and the appearance of personal computers has greatly reduced the user’s cost of use, so

that each enterprise can afford its own data center at a price. Today, the emergence of the Internet and cloud computing has made it possible to further reduce costs. If costs can be reduced, companies will of course consider adopting new technologies.

What is the trick to saving costs in cloud computing? In fact, it is the scale effect. For example, for power generation, each household uses its own generator to generate electricity. Obviously, the total cost is higher than that of centralized power supply through power plants. Another example is transportation. It is obviously more economical to use a bus to transport the same number of people than a car. Through scale, cloud computing can not only reduce fixed asset investment, but also reduce operating costs. When resources are concentrated, time-sharing or partition sharing of resources can make the same resources play a greater role. Coupled with intelligent resource allocation, the maximum use of resources can be realized. As far as energy usage efficiency is concerned, the value of Power Usage Effectiveness (PUE) has become an internationally accepted measure of data center power usage efficiency. The PUE value refers to the ratio of all energy consumed by the data center to the energy consumed by the IT load. The benchmark is 2, and the closer to 1, the better the energy efficiency level or the higher the power usage efficiency. The average PUE value of the data center is 1.21, so the use of the data center can greatly save energy.

6. Big data

Big data is another major driving force for cloud computing. Because processing massive amounts of data requires massive storage capacity and massive computing power. The general IT architecture is already incompetent, so standard equipment clusters were born, which evolved into cloud computing platforms. In fact, the two well-known commercial cloud platforms—Amazon's AWS and Google's App Engine are both spawned by processing big data.

In addition, some other driving forces that promote the development of cloud computing include the following.

- Improve resource utilization, save energy, and reduce consumption: Cloud computing (strictly speaking, virtualization) can increase server utilization from 15% to 60% or even higher, thereby reducing the energy consumption of unit computing tasks.
- Reduce the maintenance cost of the information system: The maintenance is all in one place and completed by specialized personnel.
- Improve the security posture of IT assets: All security issues are solved in one place, which is much easier than scattered in the business location of many users.
- Improve the disaster recovery capability of the information system: Cloud computing providers can conduct centralized investment and management for disaster recovery.

All in all, the driving force that promotes the emergence and development of cloud computing is economic, flexibility, convenience, elasticity, unlimited, and charge by usage.

1.4.4 The Development of Cloud Computing

Since the initial appearance of the concept of cloud computing, enterprise IT architecture has evolved from a traditional non-cloud architecture to a target cloud-based architecture. In summary, it has experienced the following three major milestone development stages.

1. Cloud computing 1.0

IT infrastructure resource virtualization stage for data center administrators. The key feature of this stage is that through the introduction of computing virtualization technology, enterprise IT applications are completely separated and decoupled from the underlying infrastructure, and multiple enterprise IT application instances and operating environments (guest operating systems) are reused in on the same physical server. And through virtualized cluster scheduling software, more IT applications are reused on fewer server nodes, thereby achieving an improvement in resource utilization efficiency.

2. Cloud computing 2.0

Resource servicing and management automation stage for infrastructure cloud tenants and cloud users. The key features of this stage are reflected in the introduction of standardized services and resource scheduling automation software on the management plane, as well as software-defined storage and software-defined network technologies on the data plane, for internal and external tenants. This would transform the complex and inefficient application, release, and configuration process of infrastructure resources that originally required manual intervention by data center administrators into one-click, fully automated resource distribution service process under necessary restricted conditions (such as resource quotas and permission approval). This change has greatly improved the rapid and agile distribution of infrastructure resources required for enterprise IT applications, shortened the preparation cycle of infrastructure resources required for enterprise IT applications to go online, and transformed the static rolling plan of enterprise infrastructure into the elastic on-demand supply of dynamic resources. This change also laid the foundation for enterprise IT to support its core business to move toward agility and better respond to the ever-changing business competition and development environment of the enterprise. In the cloud computing 2.0 stage, the provision of infrastructure resource services for cloud tenants can be in the form of a Virtual Machine (VN), a container (lightweight virtual machine), or a physical machine. The evolution of enterprise IT cloudification at this stage does not involve changes in enterprise

IT applications, middleware, and database software architectures above the infrastructure layer.

3. Cloud computing 3.0

Distributed microservices of enterprise application architecture for enterprise IT application developers and management and maintainers, Internet reconstruction of enterprise data architecture, and big data intelligence stage. The key feature of this stage is reflected in the fact that the enterprise IT's own application architecture has gradually shifted from (relying on traditional business databases and middleware business suites, specifically designed for each business application field, chimney-like, high-complexity, stateful, large-scale) vertical scale application layered architecture to (relying on open source enhanced, highly shared across different business application domains) database, middleware platform service layer and (more lightweight and decoupling functions, complete separation of data and application logic) distributed stateless architecture. This enables enterprise IT to reach a new level in supporting enterprise business agility, intelligence, and resource utilization efficiency and pave the way for the rapid iterative development of enterprise innovative business.

Regarding the above three development milestones, cloud computing 1.0 is already the past, and some industries and enterprise customers have completed the initial scale of cloud computing 2.0 construction and commercial use and are considering further expansion at this stage and the evolution toward cloud computing 3.0. Another part of the customers is moving from cloud computing 1.0 to cloud computing 2.0, and even start the evaluation and implementation of the evolution of cloud computing 2.0 and cloud computing 3.0 simultaneously.

1.5 The Advantage of Cloud Computing

The implementation and innovation of any technology is to meet the application needs of a certain group of people. Cloud computing is not an exception. It gradually penetrates into all areas of people's life and production, bringing convenience and benefits to people. The advantages of cloud computing are as follows:

1. Cut costs

Through cloud computing, companies can minimize or completely cut initial investment because they do not need to build data centers or build software/hardware platforms on their own, nor do they need to hire professionals for development, operation, and maintenance. It is usually much cheaper to use cloud computing services than to purchase software/hardware to build the required system.

2. Data can be accessed instantly anytime, anywhere

"Cloud" brings greater flexibility and mobility. Using the cloud, companies can instantly access their accounts through any device anytime, anywhere; data

can be stored, downloaded, restored, or processed easily, saving a lot of time and effort.

3. Improve adaptability and flexibly expand it needs

In most cases, the capacity of the IT system does not match the needs of the enterprise. If an enterprise configures IT equipment according to the peak demand, it will be idle at ordinary times, resulting in a waste of investment. If an enterprise configures IT equipment according to average demand, it will not be enough during peak demand. However, with cloud services, companies can have more flexible choices and can increase, decrease, or release the resources they apply for at any time.

4. Unified platform

Companies may be running different types of platforms and devices at the same time. In the cloud service platform, the application and the hardware platform are not directly related, thereby eliminating the need for multiple versions of the same application.

1.6 Classification of Cloud Computing

The layering of clouds focuses on the construction and structure of the cloud, but not all clouds of the same construction are used for the same purpose. Traditional operating systems can be divided into desktop operating systems, host operating systems, server operating systems, and mobile operating systems. Cloud platforms can also be divided into many different types. Cloud classification is mainly based on the cloud's operating mode and service mode. The former category is concerned with who owns the cloud platform, who is operating the cloud platform, and who can use the cloud platform. From this perspective, clouds can be divided into public clouds, private clouds (or dedicated clouds), community clouds, hybrid clouds, and industry clouds. The latter classification is based on the service model of cloud computing, and the cloud can be divided into three layers: IaaS, PaaS, and SaaS.

1.6.1 Classification by Operating Model

1. Public cloud

Public cloud is a type of cloud environment that can be publicly accessed, usually owned by a third-party cloud service provider. It is called public cloud because it can be accessed by the unrestricted public. Public cloud service providers can provide the installation, management, deployment, and maintenance of IT resources in all aspects, from applications and software operating environments to physical infrastructure. End-users achieve their goals through shared IT resources, and only pay for the resources they use, and obtain the IT resource services they need in this relatively economical way.

In the public cloud, users do not know with whom to share resources, and how the underlying resources are implemented, and they cannot control the physical infrastructure. Therefore, the cloud service provider must guarantee the security and reliability of the provided resources and other non-functional requirements. The level of these non-functional services also determines the service level of the cloud service provider. For those cloud services that need to strictly comply with security and regulatory compliance, higher level and more mature service levels are required. Examples of public clouds include foreign Google App Engine, Amazon EC2, IBM Developer, etc. domestic Tencent Cloud, Alibaba Cloud, Huawei Cloud, Ucloud, etc.

2. Private cloud

Enterprises and other social organizations are not open to the public. Data centers that provide cloud services (IT resources) for the enterprises or organizations are called private clouds. Compared with public clouds, users of private clouds own the entire cloud center facility, can control where program run and can decide which users are allowed to use cloud services. Since private cloud services are provided for enterprises or organizations, private cloud services can be less subject to many restrictions that must be considered in public clouds, such as bandwidth, security, and regulatory compliance. Moreover, private clouds can provide more guarantees of security and privacy through means such as user range control and network restrictions.

The types of services provided by private clouds can also be diversified. Private cloud can not only provide IT infrastructure services, but also support cloud services such as application and middleware operating environment, such as internal management information system (IMS) cloud services.

3. Community cloud

Both public and private clouds have disadvantages. A compromised cloud is the community cloud. As the name suggests, it is a cloud platform owned by a community, not an enterprise. Community cloud generally belongs to a certain enterprise group, institution alliance or industry association, and generally also serves the same group, alliance, or association. If some organizations are closely connected or have common (or similar) IT needs and trust each other, they can jointly construct and operate a community cloud in order to share infrastructure and enjoy the benefits of cloud computing. All members of the group can use the community cloud. In order to facilitate management, community cloud is generally operated and maintained by one organization, but it can also be managed by a cloud platform operation and maintenance team formed by multiple organizations.

Public cloud, private cloud, and community cloud are shown in Fig. 1.8.

4. Hybrid cloud

Hybrid cloud combines “public cloud” and “private cloud” together. Users can partly own and share partly with others in a controlled way. Enterprises can take advantage of the cost advantages of public clouds to run non-critical applications on the public cloud, and at the same time provide services through the internal

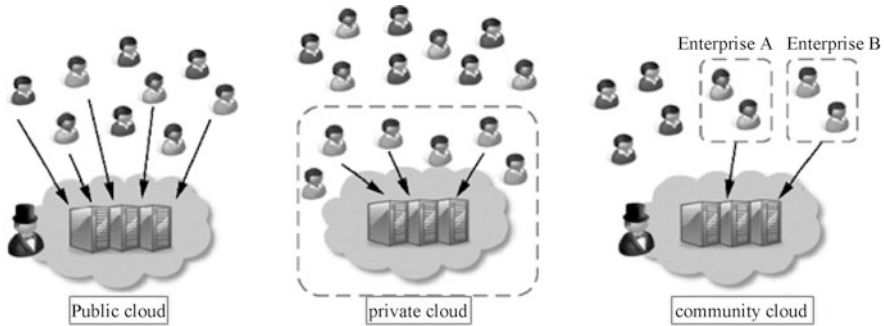


Fig. 1.8 Public cloud, private cloud, and community cloud

private cloud for major applications with higher security requirements and more criticality.

There are many reasons for using hybrid cloud. There are two main reasons: the compromise of various considerations; the transition from private cloud to public cloud. For the first reason, although some organizations are eager to use the public cloud, because of various regulations, confidentiality requirements or security restrictions, they cannot put all their resources on the public cloud, so some IT resources will be deployed in the public cloud. In the above situation, part of the IT resources is deployed in the business location, which will form a hybrid cloud.

In the long run, public cloud is the mainstream of cloud computing development due to its higher resource utilization efficiency, but private cloud and public cloud will coexist for a long time in the form of common development. Just like the emergence of banking services, the transfer of currency from individuals to bank custody is a safer and more convenient process, but some people may choose to keep them on their own.

5. Industry cloud

The industry cloud is for the purpose of the cloud, not for the owner or user of the cloud. If the cloud platform is customized for a certain industry (e.g., for the automotive industry), it is called an industry cloud. The components used in the industry cloud ecological environment should be more suitable for related industries, and the software deployed on it is also industry software or its supporting software. For example, for the cloud platform established by the hospital, the data storage mechanism deployed above should be particularly suitable for the storage, indexing, and query of medical data.

There is no doubt that the industry cloud is suitable for the specified industry, but it may be of little value to the average user. Generally speaking, the structure of the industry cloud will be simpler, and its management is usually taken care of by the industry's "leading" or a computing center (supercomputer center) designated by the government.

The relationship between the industry cloud and the four types of clouds mentioned above is not exclusive, and there may be an overlapping or overlapping relationship between them. For example, industry clouds can be built on public clouds, private clouds, and more likely community clouds.

6. Other cloud types

In addition to the cloud types above, there are other cloud types. For example, according to whether the cloud is aimed at individuals or enterprises, it can be divided into consumer cloud and enterprise cloud. The consumer cloud audience is the general public or individuals, so it is also called the public cloud. This kind of cloud promotes personal storage and document management needs; the enterprise cloud is for enterprises and promotes comprehensive IT services for enterprises. The classification of these clouds is still a certain segmentation or combination of the above cloud types in essence.

1.6.2 *Classification by Service Model*

According to the service model of cloud computing, the cloud can also be divided into three layers: IaaS, PaaS, and SaaS. Different cloud layers provide different cloud services. Figure 1.9 shows the composition of a typical cloud computing.

1. IaaS

IaaS is at the bottom of the three-layer service of cloud computing, and it is also the scope covered by the narrow definition of cloud computing. IaaS provides IT infrastructure to users in the form of services like water and electricity and provides highly scalable and on-demand IT capabilities based on hardware resources such as servers and storage in the form of services. It is usually charged according to the cost of the resources consumed.

This layer provides basic computing and storage capabilities. Taking the provision of computing capabilities as an example, the basic unit it provides is a virtual server, including CPU, memory, operating system, and some software, as shown in Fig. 1.10. Specific instance is Amazon EC2.

2. PaaS

PaaS is located in the middle of the three-layer service of cloud computing and is often referred to as a “cloud operating system,” as shown in Fig. 1.11. It provides end-users with an Internet-based application development environment, including application programming interfaces and operating platforms and supports various software/hardware resources and tools required for the entire life cycle of applications from creation to operation. The billing is usually based on user or login status. At the PaaS layer, service providers provide encapsulated IT capabilities, or some logical resources, such as databases, file systems, and application operating environments. Examples of PaaS products include Huawei’s software development cloud DevCloud, Salesforce’s [Force.com](https://www.salesforce.com), and Google’s Google App Engine.

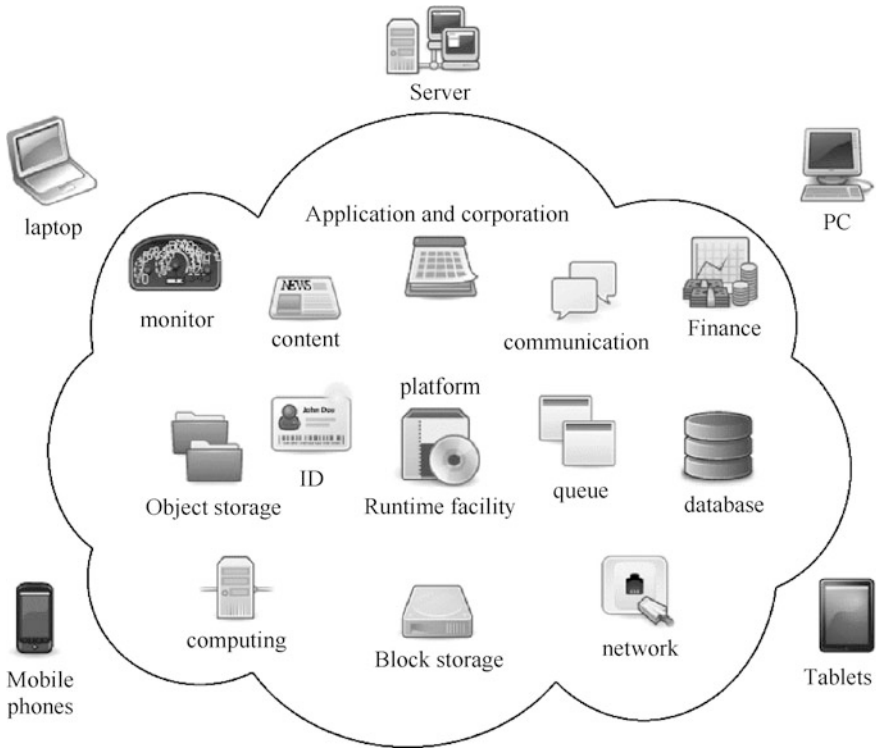


Fig. 1.9 Components of cloud computing

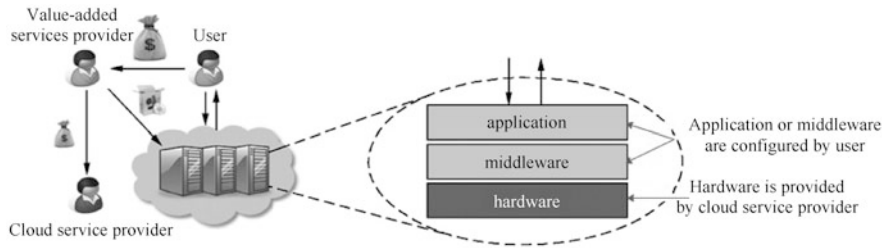


Fig. 1.10 IaaS structure

PaaS is mainly for software developers. It used to be a difficult problem for developers to write and run programs in a cloud computing environment through the network. Under the premise of gradual increase in network bandwidth, the emergence of two technologies has solved this problem. One is online development tools. Developers can use browsers, remote consoles (running development tools in the console), and other technologies to directly develop applications remotely, without the need to install development tools locally; the other is

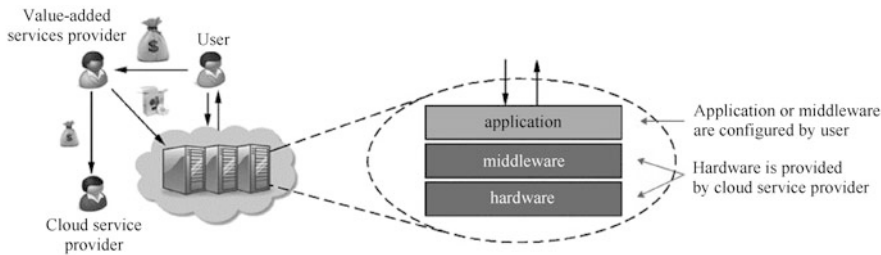


Fig. 1.11 PaaS structure

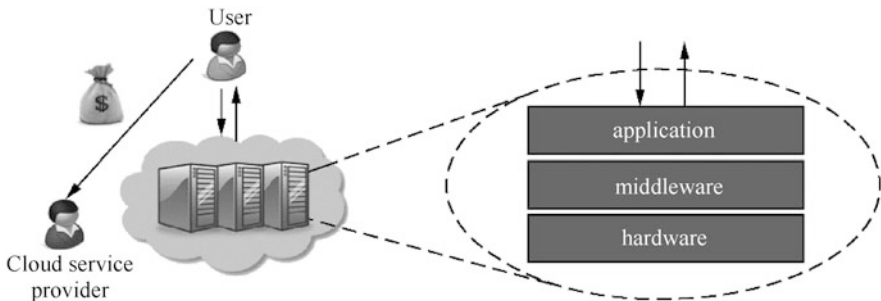


Fig. 1.12 SaaS structure

local development tools and cloud computing integrated technology, that is, deploying the developed application to the cloud computing environment through local development tools, while enabling remote debugging.

3. SaaS

SaaS is the most common cloud computing service, located at the top of the three-tier cloud computing service, as shown in Fig. 1.12. The user uses the software on the Internet through a standard Web browser. Cloud service providers are responsible for maintaining and managing software and hardware facilities and provide services to end-users for free or on-demand rental.

These services are both for general users, such as Google Calendar and Gmail, and for enterprise groups to help with payroll processes, human resource management, collaboration, customer relationship management and business partner relationship management, such as [Salesforce.com](https://www.salesforce.com) and Sugar CRM. These SaaS-provided applications reduce the time for users to install and maintain software and their skills requirements and can reduce software license fees through pay-per-use.

The above three layers, each has corresponding technical support to provide the services of this layer, with the characteristics of cloud computing, such as elastic scaling and automatic deployment. Each layer of cloud services can be independent into a cloud or based on the services provided by the clouds below. Each kind of

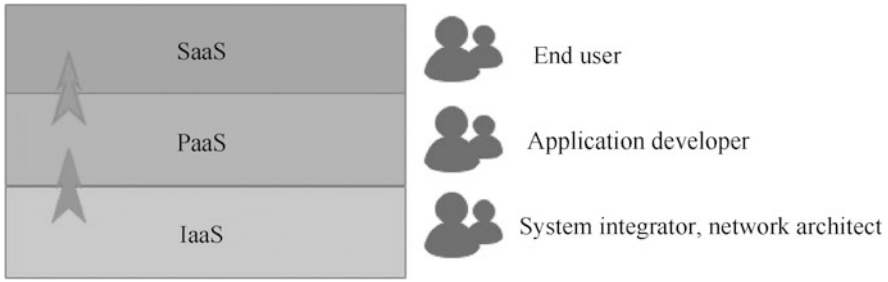


Fig. 1.13 Types of user groups of the three types of service models

cloud can be directly provided to end-users for use, or it can only be used to support upper-layer services. The three types of service models usually have different user groups (see Fig. 1.13).

1.7 Cloud Enabling Technology

This section introduces some basic key technologies included in the current cloud computing technology, which called cloud enabling technology. This includes broadband network and internet architecture, data center technology, virtualization technology, Web technology, multi-tenant technology, and service technology.

From a technical point of view, cloud computing is inextricably linked with various technologies such as distributed systems, virtualization technology, and load balancing. Like the “Je Kune Do” in information technology although it combines the essence of various types, it still forms its own types.

In terms of specific technical realization, the cloud platform innovatively integrates a variety of technical ideas, through different combinations, to solve different problems encountered in specific applications. Therefore, people will find a variety of technologies in the cloud platform, and some people will also judge that cloud computing is nothing but an old tune. However, if we only focus on the existence of a certain technology and ignore the integration and innovation of cloud computing itself in technical applications, there will be a situation of “seeing the trees, not the forest,” which is not only biased, but also leads to perception errors.

As far as technology is concerned, cloud computing is essentially derived from ultra-large-scale distributed computing and is an evolved distributed computing technology. Cloud computing also extends the Service-Oriented Architecture (SOA) concept and integrates virtualization, load balancing, and other technologies to form a new set of technical concepts and implementation mechanisms. Specifically, the core significance of cloud computing lies not only in the development of technology, but also in the organization of various technologies to change people’s

thinking about building IT systems and at the same time make fundamental changes in the structure.

1.7.1 Broadband Network and Internet Architecture

All clouds must be connected to the network, and this inevitable requirement forms an inherent dependence on network interconnection. The Internet allows remote provision of IT resources and directly supports ubiquitous network access. Although most clouds rely on the Internet for access, cloud users can also choose to access the cloud only through private or proprietary network connections. The attractiveness of the cloud platform is closely related to the quality of the service provided by the access network.

The Internet's largest backbone network is established and deployed by ISPs, and they rely on core routers for strategic interconnection. These routers are in turn connected to transnational networks in the world. Figure 1.14 shows the interconnection between a backbone ISP network and other ISP networks and various organizations.

The concept of the Internet is based on a non-centralized supply and management model. ISPs can freely deploy, operate, and manage their networks, and they can also freely choose other ISPs that need to be interconnected. Although there are companies such as The Internet Corporation for Assigned Names and Numbers (ICANN) that supervise and coordinate important Internet affairs, there is actually no central entity to fully control the Internet. For specific countries, government regulations and regulatory laws supervise the services provided by domestic institutions and ISPs.

The topology of the Internet has become a dynamic and complex collection of ISPs. These ISPs are highly interconnected through their core protocols. Smaller branches are expanded from the main node, and these branches can be extended outward to new branches, until finally reaching every digital device or terminal connected to the Internet.

The communication path connecting cloud users and their service providers may include multiple ISP networks. The net structure of the Internet connects the digital terminals that are connected through a number of selectable network routes. In actual operation, it is determined which route to choose according to the current network conditions. Therefore, when using cloud services, even if a certain network or even multiple networks fails, the communication is generally guaranteed to be uninterrupted, but it may cause routing fluctuations or delays.

Some more detailed network knowledge, we will specifically introduce in Chap. 4.

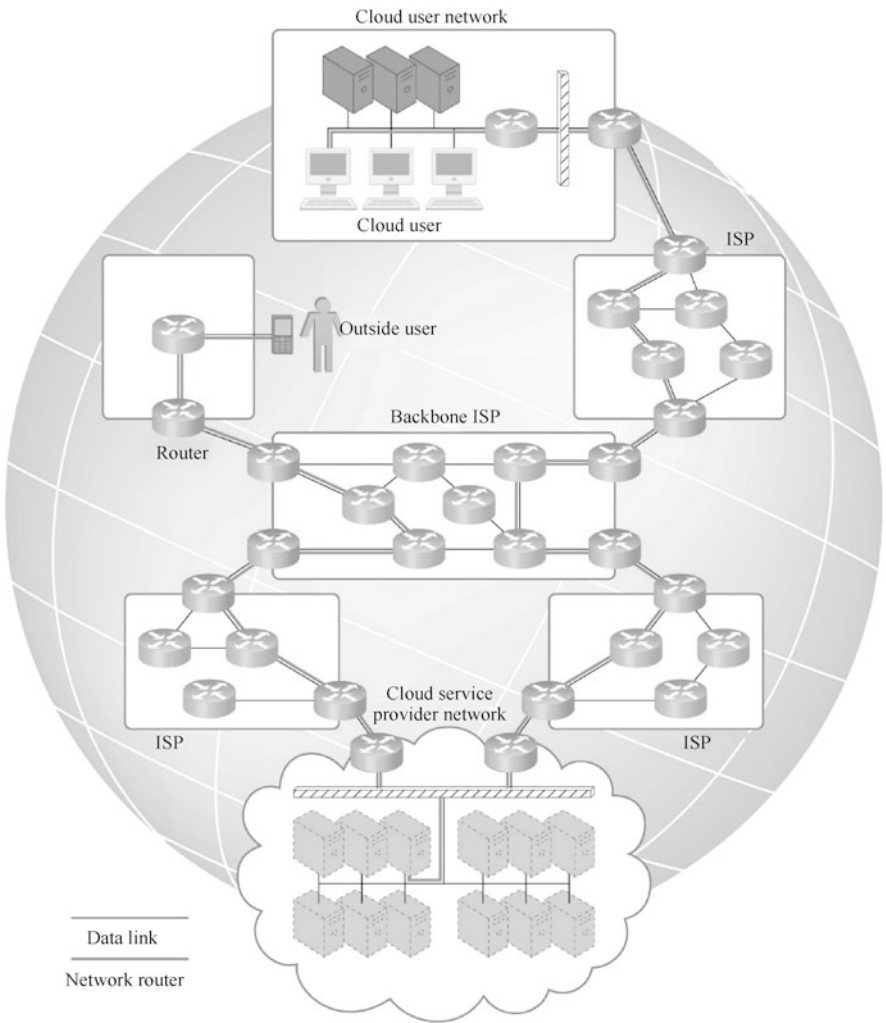


Fig. 1.14 The network of a backbone ISP is interconnected with the networks of other ISPs and various organizations

1.7.2 Data Center Technology

Compared with geographically dispersed IT resources, effectively organizing many IT resources adjacent to each other to form a data center is more conducive to energy sharing, increasing the utilization rate of shared IT resources, and improving IT personnel’s efficiency. These advantages make data centers bloom all over the world. A modern data center refers to a particular type of IT infrastructure used to

centrally place IT resources, including servers, databases, network and communication equipment, and software systems.

The data center contains physical and virtual IT resources. The physical layer refers to the infrastructure where computing/networking systems and equipment, hardware systems, and operating systems are placed. The virtualization layer abstracts and controls resources and is usually composed of operation and management tools on the virtualization platform. The virtualization platform outlines physical computing and network IT resources into virtualized components, which makes it easier to allocate, operate, release, monitor, and control resources.

The data center is based on standardized commercial hardware, designed with a modular architecture, and integrates multiple identical infrastructure modules and equipment, with features such as scalability, substitutability, and the ability to quickly replace hardware. Modularity and standardization are critical conditions for reducing investment and operating costs because they can achieve economies of scale in procurement, deployment, operation, and maintenance.

Common virtualization strategies and the ever-improving capacity and performance of physical devices have promoted IT resources because fewer physical components can support more complex configurations. Integrated IT resources can serve different systems and can also be shared by other cloud users.

Data centers usually have the following characteristics.

1. Automation

The data center has a unique platform that can automate tasks such as provisioning, configuration, patching, and monitoring without manual operations.

2. Remote operation and management

In the data center, most of the IT resources' operation and management tasks are completed through the network remote console and management system. Technicians generally do not need to enter the dedicated room where the server is placed unless they perform equipment handling, wiring, or hardware-level installation and maintenance tasks.

3. High availability

For data center users, any form of downtime in the data center will significantly impact the continuity of their tasks. Therefore, to maintain high availability, data centers have adopted increasingly high redundancy designs; to cope with system failures, data centers usually have redundant uninterruptible power supplies, integrated wiring, and environmental control subsystems; for load balancing, then there are redundant communication links and cluster hardware.

1.7.3 Virtualization Technology

Among the many existing definitions of cloud computing, there is a definition that describes cloud computing as “accessible through the network, accessible on-demand, subscription-paid, shared by others, packaged outside of your own

data center, simple Easy-to-use, virtualized IT resources.” Although this definition is not comprehensive, it at least points out that virtualization technology is essential for cloud computing.

Virtualization is a virtual (rather than real) version created for certain things, such as hardware platforms, computer systems, storage devices, and network resources. Its purpose is to get rid of the various limitations of physical resources in reality, that is, “virtualization is a logical representation of resources, and physical limitations do not restrict it.”

Although many people are interested in virtualization technology because of cloud computing, virtualization technology is not a new technology. From the virtualization of IBM’s mainframe computers to the current VMware series of desktops by EMC (Ian Xin company, acquired VMware in 2003), stand-alone virtualization technology has experienced more than half a century of development. In the early days, virtualization technology was implemented to make a single computer look like multiple computers or completely different computers, thereby improving resource utilization and reducing IT costs. With the development of virtualization technology, the scope of the concept of virtualization is also increasing.

Computer systems are usually divided into several levels, from bottom to top, including the underlying hardware resources, operating systems, user software, etc. The emergence and development of virtualization technology enable people to abstract various underlying resources to form different “virtual layers” and provide upward with the same or similar functions as the real “layers,” thereby shielding the differences in equipment and making the underlying equipment Transparent to upper-level applications. Virtualization technology reduces the degree of coupling between resource users and resource entities so that users no longer depend on specific types of certain types of resources.

The virtualization involved in cloud computing is a higher level of virtualization after development. It means that all resources-computing, storage, applications, and network equipment are connected and managed, and scheduled by the cloud platform. With the help of virtualization technology, the cloud platform can uniformly manage the diverse resources at the bottom. It can also conveniently manage resource scheduling at any time and realize the on-demand allocation of resources so that a large number of physically distributed computing resources can be logically controlled. It is presented in an overall form and supports various application requirements. Therefore, the development of virtualization technology is a crucial driving force for cloud platforms.

Although virtualization is a crucial component of cloud computing, cloud computing is not limited to virtualization. Cloud computing also expresses the service model of on-demand supply and billing and technical characteristics such as flexibility, transparency, and building blocks. Chapter 3 of this book will describe in detail the related technologies of virtualization.

1.7.4 Web Technology

Cloud computing has a deep-rooted dependence on the Internet and Web technologies. Web technology is often used as the realization medium and management interface of cloud services.

1. Basic web technology

The World Wide Web is a system of interconnected IT resources accessed through the Internet. Its two fundamental components are the Web browser client and the Web server. Other components, such as proxies, caching services, gateways, and load balancing, are used to improve Web application features such as scalability and security. These additional components are located in the hierarchy between the client and the server.

Web technology architecture consists of three basic elements.

- Uniform Resource Locator (URL): A standard syntax used to point to Web resources' identifier. URLs usually consist of logical network locations. If you want to locate Huawei's official website, enter the URL in the browser, and you can see the homepage of the official website.
- HyperText Transfer Protocol (HTTP): The basic communication protocol for exchanging content and data through the World Wide Web. Usually, the URL is transmitted via HTTP. When a user visits a webpage, the browser will send a webpage request to the website corresponding to the URL, using HTTP, and the website will return the requested webpage to the browser after receiving the request.
- Markup Language-Markup Language: It provides a lightweight method to represent Web-centric data and Metadata. At present, Hyper Text Markup Language (HTML) is commonly used in webpages, and the meaning of its tags is fixed. For example, the tag `<p>` means segmentation in the page; while the user defines Extensible Markup Language (XML) tags, the user can freely assign meaning to the Web data through the metadata. Metadata refers to data describing other data, or structural data used to provide information about a certain resource. If "`<author>Lu Xun</author>`" is defined in the XML file, the custom tag `<author>` here belongs to metadata. HTML and XML are the two main markup languages that are currently widely used.

Web resources are also called hypermedia to distinguish them from hypertext. This also means that all kinds of media, such as images, audio, video, and plain text, can be referenced in a single file. However, some types of hypermedia require additional software or Web browser plug-ins to play or watch. The Web browser can request to perform read, write, update, or delete operations on Web resources on the Internet, and identify and locate them through the URL of the resource. A request for a webpage is sent to a resource host identified by a URL through HTTP, and then the Web server locates the resource and processes the requested operation, and sends the processing

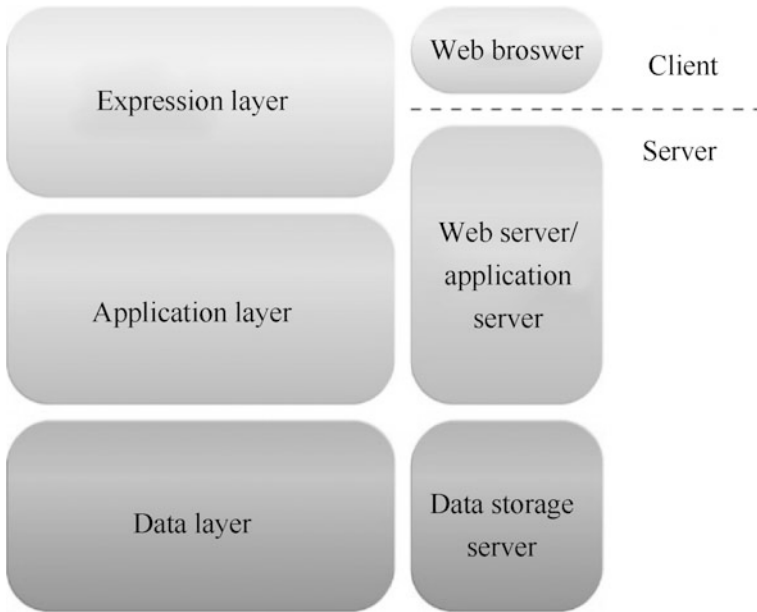


Fig. 1.15 The three-layer model of Web applications

result back to the browser client. Processing results generally consist of HTML or XML statements.

2. Web application

Distributed applications based on Web technology (usually displaying the user interface through a Web browser) are generally considered Web applications. Due to their high accessibility, these applications appear in all types of cloud-based environments.

A typical Web application may have a three-tier model, as shown in Fig. 1.15. The first layer is the Presentation Layer, which is used to represent the user interface. The second layer is the Application Layer, which is used to implement application logic. The third layer is the Data Layer, which consists of persistent data storage. This pattern is also commonly referred to as the Model-View-Controller (MVC) pattern.

The MVC pattern model is the part of the application used to process the data logic of the application. Usually, model objects are responsible for accessing data in the database. This is corresponding to the data layer. The core device is a Data Storage Server or a Database Server. The view is the part of the application that handles the display of data. Usually, views are created based on model data. This corresponds to the presentation layer. The browser on the client-side is used for requesting and displaying Web data; the Web server on the server-side is used

to process the browser's request and return the requested webpage (corresponding to a static website) or corresponding to the results of program execution (corresponding to dynamic webpages). The controller is the part of the application that handles user interaction. Usually, the controller is responsible for reading data from the view, controlling user input, and sending data to the model. Corresponding to the application layer here, it mainly deals with transaction logic, and the core device is the Application Server.

The MVC model helps manage complex applications, simplifies group development, and makes application testing easier.

1.7.5 Multi-Tenant Technology

- The purpose of designing multi-tenant applications is to make it possible for multiple users (tenants) to logically access the same application simultaneously. Each user has his view of the application that he uses, manages, and customizes, corresponding to a specific instance of the application. At the same time, each tenant will not realize that other tenants are using the application.

Multi-tenant applications ensure that each tenant will not access data and configuration information that is not their own. Moreover, each tenant can independently customize its application features.

- User Interface: Tenants can define application interfaces with special interface appearances.
- Business Process: When implementing applications, tenants can customize the rules, logic, and workflow of business processing.
- Data Model: Tenants can extend the application data model to include, exclude or rename the fields of the application data structure.
- Access Control: Tenants can independently control the access rights of users or groups.

Multi-tenant application architecture is usually much more complicated than single-tenant application architecture. It needs to support multi-user sharing of various components (including entrance, data model, middleware, and database), and it also needs to maintain a security level to isolate different tenants' operating environments.

The general characteristics of multi-tenant applications are as follows:

- Usage Isolation: The usage behavior of a tenant will not affect the availability and performance of the application to other tenants.
- Data Security: Tenants cannot visit data of other tenants.
- Recovery: Data backup and recovery processes are operated separately for each tenant.

- **Application Upgrade:** The upgrade of shared software will not have negative influence on tenants.
- **Scalability:** The application can be expanded according to the increase in the use demand of existing tenants or the increase in the number of tenants.
- **Metered Usage:** Charge based on the application processing and functions actually used by the tenant.
- **Data Tier Isolation:** Tenants have independent databases, tables, and schemas that are isolated from other tenants. Or, it can also be specially designed for multi-tenant shared databases, tables, and schemas.

1.7.6 Service Technology

Service technology is the foundation of cloud computing. It laid the foundation of “as a service” cloud delivery model. The typical realization and construction of service technology of cloud computing are as follows:

1. Web service

The technical standard of Web service are as follows:

- **Web Service Description Language, WSDL:** This markup language is used to create WSDL definitions, which define the application programming interface (API) of Web services, including its independent operations (functions) and input/output messages for each operation.
- **XML Schema Definition Language, XML:** The messages exchanged by Web services are generally expressed in XML. This language is used to describe the XML schema. The XML schema defines the data structure of XML-based input/output messages, which are exchanged by Web services. The XML schema can be directly linked to the WSDL definition or embedded in the WSDL definition.
- **Simple Object Access Protocol, SOAP:** This protocol defines the general message format of the request and response messages exchanged by Web services. A SOAP message consists of a body and a header. The body is the content of the message. The header generally contains metadata that can be processed at runtime.
- **Universal Description, Discovery, and Integration, UDDI protocol:** The protocol stipulates that the server must register and publish the WSDL definition to the service catalog so that users can discover the service.

The above four technologies form a classic Web service technology, and the relationship is shown in Fig. 1.16.

2. REST service

Compared with complex technologies such as SOAP and WSDL, REST (Representational State Transfer translated as representational state transfer) is a

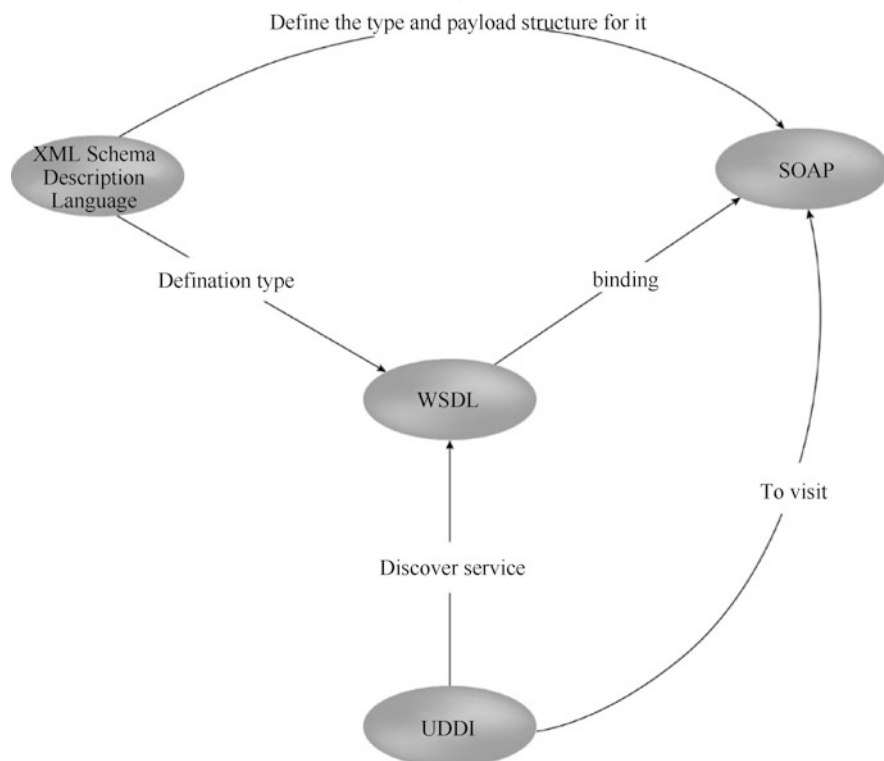


Fig. 1.16 The relationship of four Web service technologies

lightweight and concise software architecture style that can reduce the complexity of development and improve the scalability of the system. Mainstream cloud service providers are increasingly adopting REST-style design and implementation to provide Web services. For example, Amazon offers REST-style Web services for book searches; Yahoo! provides REST-style Web services.

REST services do not have an independent technical interface. Instead, they share a common technical interface called a uniform contract, which corresponds to a set of architectural constraints and principles. An application or design that meets these constraints and regulations is called REST. REST is usually based on existing widely popular protocols and standards such as HTTP, Uniform Resource Identifier (URI), XML, and HTML.

3. Service agent

Cloud-based environments rely heavily on service agents to perform most tasks such as monitoring and metering. These service agents are usually customized to accomplish specific tasks such as elastic expansion and pay-per-use. The service agent is an event-driven program, which intercepts messages and performs related processing at runtime.

Service agents are divided into active service agents and passive service agents. Both service agents are common in cloud-based environments. After the active service agent intercepts and reads the message, it may take certain measures, such as modifying the message content or message path. The passive service agent does not modify the content of the message, but after reading the message, it captures the specific content for monitoring, recording, or reporting.

1.8 Understand Cloud Computing

We can understand cloud computing from multiple perspectives: cloud computing is a business model and a computing model or an implementation method. At the same time, open source technology has also been widely used in cloud computing. Open source technology and software have promoted the development of cloud computing technology, and the research on cloud computing technology has further boosted the development of open source technology. The two complement each other.

1.8.1 *The Ternary Epistemology of Cloud Computing*

With the development of cloud computing ecosystem, today's cloud computing should include three aspects: business model, computing model, and implementation method.

1. Cloud computing as a business model

Cloud computing services represent a new business model. SaaS, PaaS, and IaaS are the three manifestations of this business model. For any business model and being feasible in theory, it is also necessary to ensure that it is feasible in practice. Therefore, along with the development of cloud computing service concepts, cloud computing has also formed a set of software architecture and technical implementation mechanisms, and the cloud platform we often hear is a concrete manifestation of this mechanism.

Amazon sells all “commodities” suitable for e-commerce, including books, DVDs, computers, software, video games, electronic products, clothes, furniture, computing resources, and so on. When launching EC2, Amazon also faced many questions about “why does this retailer want to do this,” but the company's CEO Bezos had a much broader understanding of business concepts at the time. Bezos believes that whether it is “personal computer + software” or this way of obtaining services from the “cloud,” it is not only a technical issue, but also a “business model.”

To enable the website to support large-scale business, Amazon has made excellent infrastructure construction efforts and has naturally accumulated much experience. In order to sell a large number of idle computing resources as

commodities, Amazon has successively launched storage and computing rental services such as S3 (Simple Storage Service) and EC2. Bezos said, “We think this will be an exciting business someday, so our purpose of doing this is straightforward: we think this is a good business.” Although the media thinks this is a safe time for Bezos. A risky bet after the dot-com bubble, “The CEO of Amazon wants to use the technology behind his website to run your business, but Wall Street only wants him to be optimistic about his storefront.” But EC2 did affect the entire industry, and it also affected many people, the industry was obviously shaken at the time.

Before Amazon, although many services have the characteristics of cloud computing services, even the services provided by Google can still be regarded as a business model within the meaning of Internet services. After Amazon launched IaaS, it seems to have opened a window to the Internet world, telling people that computing resources can also be operated in this way, and there is a new business model called cloud computing. And those service models that are similar to traditional Internet services can finally be independent and find their own position-cloud computing services.

The University of California, Berkeley pointed out in a report on cloud computing that cloud computing refers to applications provided in the form of services on the Internet and the hardware and software that provide these services in data centers. The hardware and software are called “clouds.”

The National Institute of Standards and Technology issued Cloud Computing Synopsis Recommendations in 2011, in which PaaS, SaaS, and IaaS were explained in detail. Many people think that SaaS must run on PaaS, PaaS must run on IaaS, but in fact, there is no absolute hierarchical relationship between the three. They are all a kind of service, which can have a hierarchical stacking relationship or not.

2. Cloud computing as a computing model

From the perspective of computing mode, the earliest origin of cloud computing should be ultra-large-scale distributed computing. For example, Yahoo!’s ultra-large-scale distributed system designed to solve system support for large-scale applications is to decompose large problems and solve them together by a large number of computers distributed in different physical locations. However, with the continuous development and improvement of technology, cloud computing draws on many other technologies and ideas, including virtualization technology and SOA concepts, when solving specific problems. Cloud computing is fundamentally different from these technologies, not only in commercial applications but also in implementation details.

As a computing model, cloud computing has its computing boundary determined by upper-level economic and lower level technical factors. Economic factors determine the business form of this computing model from top to bottom, and technical factors determine the technical form of this computing model from bottom to top.

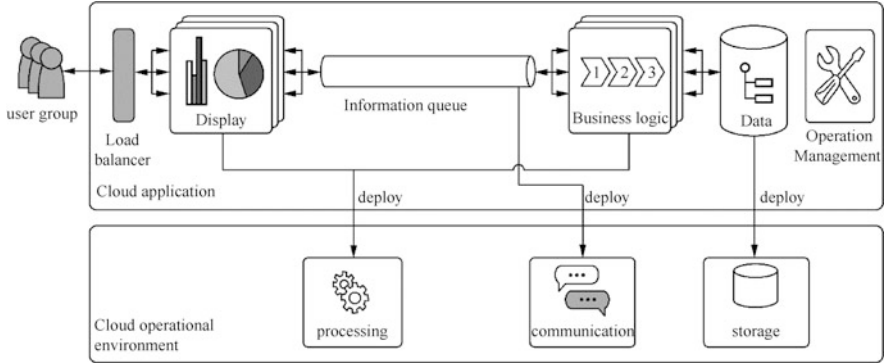


Fig. 1.17 Cloud computing’s logical structure of horizontal cloud body

The computing model as a cloud computing service can be further understood from two perspectives: the horizontal cloud body’s logical structure and the logical structure of the vertical cloud stack.

(a) Logical structure of horizontal cloud

The logical structure of the horizontal cloud body of cloud computing is shown in Fig. 1.17. From the perspective of the horizontal cloud, cloud computing is divided into two parts: Cloud Runtime Environment and Cloud Application.

The cloud runtime environment includes Processing, Communication, and Storage, which together support all aspects of upper cloud applications.

From this perspective, we can see that the structure of cloud computing is very similar to the structure of the personal computer we usually use. And Chaps. 3 to 5 of the book cover the 3 dimensions of processing (virtualization technology), communication (networking) and storage (distributed storage) respectively.

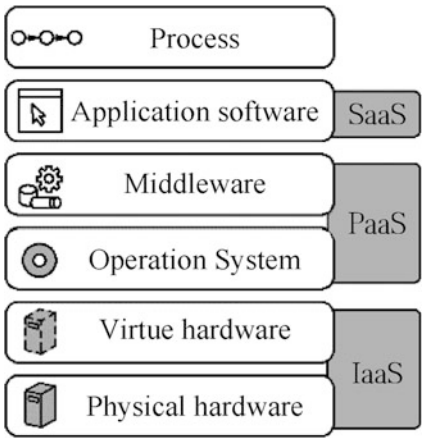
(b) Logical structure of vertical cloud stack

The logical structure of the vertical cloud stack is similar to the previous business model, and it is also composed of three parts: SaaS, PaaS, and IaaS, except that it will be viewed from a technical point of view.

SaaS, PaaS, and IaaS have become the “recognition cards” for people to perceive cloud computing. Many people will view the relationship between these three technical layers in a hierarchical manner. For example, SaaS runs on PaaS, and PaaS runs on Above IaaS. It can be further seen that the IaaS layer includes Physical Hardware and Virtual Hardware; the PaaS layer includes operating systems and middleware; and there are business processes on top of the application software of the SaaS layer. The logical structure of the vertical cloud stack of cloud computing is shown in Fig. 1.18.

From a technical point of view, there is no obvious difference between SaaS users and the users of ordinary stand-alone software. PaaS provides

Fig. 1.18 Logical structure of vertical cloud stack for cloud computing



platform services, so users are developers and need to understand the development and deployment of applications in the platform’s environment. And IaaS delivers the lowest level of infrastructure services, so the users it faces are IT managers, that is, IT managers will configure and manage them first, and then perform software deployment and other tasks on it.

Although people are accustomed to dividing services according to the content provided by service providers, there is no absolute clear boundary between these three service models. Some more powerful cloud computing service providers may provide products with both SaaS and PaaS features, and some cloud computing service providers try to provide a complete set of cloud computing services, further blurring the differences in the three service models at levels.

People are slowly realizing that there are infinite possibilities of services provided through the Internet, and many companies have discovered new directions for Internet services. Therefore, in addition to SaaS, PaaS, and IaaS, some new service form names have appeared, such as Business Process as a Service, Database as a Service, and Security as a Service.

It is undeniable that these emerging cloud computing services extend the concept of Internet services and provide information services that are more in line with the laws of commercial development. If the emergence of the Internet has greatly satisfied people’s needs for rapid acquisition and sharing of knowledge, then cloud computing services have met people’s needs for convenient acquisition, sharing, and innovation of knowledge to a greater extent based on traditional Internet services.

After this “business model” concept of “simpler, more convenient, and lower cost” through the Internet has been widely used to meet various needs, service providers are gradually trying all services that can be provided to users through the Internet. “Cloud,” so now there is a term XaaS. Where X refers to Anything or Everything, which stands for “everything can be a

service.” It now appears that the commercial practice of various new possibilities has continued to develop and enrich the possible meaning of cloud computing services.

(c) Cloud computing as a realization model

The ultimate realization of cloud computing requires a new generation of software/hardware technology to promote, that is, the current popular data center, and evolve toward a software-defined data center (SDDC). The data center is the ultimate home of cloud computing, including a full range of computing, storage, and communication requirements. With the data center’s operation, everyone began to encounter a series of common problems, including hardware resource utilization, scalability, and automated management. Hardware upgrades take years and months, and it is usually difficult to meet the needs of a fast-developing business. Software definition is a realistic and feasible way out. Therefore, the software-defined data center has quickly become a hot keyword in the IT industry.

The software-defined data center is a relatively new concept that extends virtualization concepts (such as abstraction, centralization, and automation) to all data center resources and services to achieve IT as a Service (ITaaS). In a software-defined data center, all infrastructure elements (network, storage, CPU, and security) are virtualized and delivered as services.

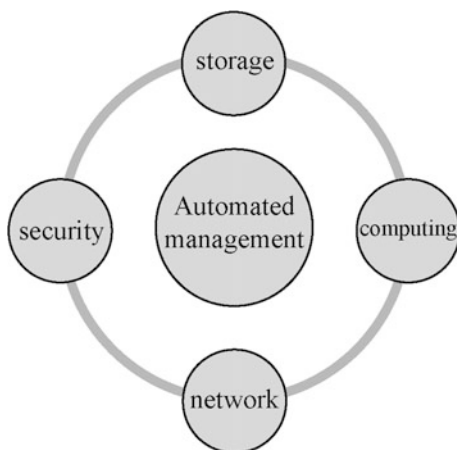
The core resources of a software-defined data center are computing, storage, and networking. These three are undoubtedly the basic functional modules. Unlike traditional concepts, the software-defined data center emphasizes the capabilities abstracted from the hardware rather than the hardware itself.

For computing, computing capabilities need to be abstracted from the hardware platform, so that computing resources can break away from the hardware constraints and form a resource pool. Computing resources also need to be able to migrate within the software-defined data center’s scope to adjust the load dynamically. Although virtualization is not a necessary condition, it is non-virtualization that can meet these requirements. The requirement for storage and network is the separation of the control plane and data plane. This is the first step to break away from hardware control, and it is also the initial stage of being able to define the behavior of these devices with software. After that, it is possible to consider connecting the control layer and the data layer to the software-defined data center. Security has increasingly become a factor that needs to be considered separately in data centers. Security hazards may appear between basic computing, storage, and the network or hidden in the data center’s management system or the user’s software. Therefore, it is necessary to regard safety as a basic functional module alone, parallel with the above three basic available modules.

Having these basic functional modules is not enough. A centralized management platform is needed to link them together, as shown in Fig. 1.19.

Automated management is the key to organizing the basic functional modules of the software-defined data center. Here it must emphasize

Fig. 1.19 Software-defined data center function division



“automated” management, not just a set of exquisite interfaces. An important driving force of the software-defined data center is the user’s management of ultra-large-scale data centers, and “automation” is undoubtedly a must.

In summary, there is no inevitable relationship between cloud computing services, cloud computing models, and cloud computing implementation. If a service implemented with traditional underlying architecture or similar to supercomputing has the three characteristics of cloud computing services: a large user base, always online, and accessibility anytime, anywhere, it can also be called cloud computing. The architecture and specific implementation itself are designed to put forward various solutions to the problems of “big users,” “big data,” and “big systems,” which are also typical problems encountered when providing cloud computing services. Therefore, cloud computing services supported by cloud computing architecture and implementation can improve service efficiency and give full play to the capabilities and advantages of cloud computing.

Just like the evolution of species, society itself will continue to advance and develop, and as a result, different progressive service models and technical needs will be generated. People’s demand for computing has promoted the popularization and development of computers, and the demand for communication and sharing has promoted the birth of the Internet. Cloud computing is also a result of social demand. With the desire for knowledge, continuous innovation and sharing, people continue to put forward new information services and products. The emergence of cloud computing, on the one hand, solves the increasingly prominent pressure problem at the system level; on the other hand, it broadens the scope of network applications and the possibility of innovation and further satisfies the premise of significantly reducing the cost of people creating and sharing knowledge. The needs of human society to acquire, innovate, and share knowledge. Therefore, cloud computing is an inevitable product of the development of the information

society. With the development of the application environment, cloud computing will become more and more popular, which will bring a new experience of information society to humanity.

One of the significances of the “industrial revolution” is to free people from the shackles of production conditions and greatly liberate the productivity of material products and tangible services. The emergence of cloud computing is also gradually freeing people from the constraints of using computing resources and information services, reducing the cost of knowledge acquisition, making knowledge generation easier and sharing more convenient. It revolutionized the productivity of information products and knowledge services. Therefore, cloud computing is as important as steam engines, internal combustion engines, and electricity and will bring about an industrial revolution in the information society.

Nowadays, cloud computing is still developing, and the extent to which it will develop in the future is still unknown. We are still exploring and deepening our understanding of cloud computing. After all, we have a process of “hearing and knowing” for new things. For cloud computing at this stage, what is most needed is support, and what is most feared is to belittle, or to be contemptuous of conclusions. But in any case, cloud computing has already had a positive impact on some areas of human society’s production and life. I believe that with the development of technology and service innovation, the “cloud computing era” will come soon, and ultimately affect each of us.

1.8.2 Open Source Methodology of Cloud Computing

Open source technology has been widely used in the field of cloud computing. In the era of cloud computing, open source is not only an open source software product, but also a methodology and a collaborative way to construct large-scale and complex software.

1. The definition of open source and relative concepts

Open source refers to the opening of the source code, source data, source assets of a type of technology or a product, which can be technologies or products of various industries, and its scope covers multiple social dimensions such as culture, industry, law, and technology. If it is software code that is open, it is generally called open source software, or open source software for short. Open source’s essence is to share assets or resources (technology), expand social value, improve economic efficiency, and reduce transaction barriers and social gaps. Open source is closely related to open standards and open platforms.

Open source software is a kind of computer software that the copyright holder provides anyone with the right to study, modify, and distribute, and publish the source code. The Open Source Initiative (OSI) has a clear definition of open

source software, and it is recognized in the industry that only software that meets this definition can be called open source software. This name stems from the proposal of Eric Raymond. OSI defines the characteristics of open source software as follows:

- The license of open source software should not restrict any individual or group from selling or gifting broad-based works containing the open source software.
- The program of open source software must contain source code, and the release of source code and subsequent programs must be allowed.
- Open source software licenses must allow modification and derivative works and be published using the original software's license terms.

An open source license is a license for computer software and other products that allows source code, blueprints, or designs to be used, modified, or shared under defined terms and conditions. At present, there are 74 kinds of OSI-certified open source licenses, and the most important are only 6–10 kinds (the most important two are GPL and Apache). Under the wave of open source commercialization, moderately loose Apache and other licenses are more popular.

Free software is software that users can freely run, copy, distribute, learn, modify, and improve. Free software needs to have the following characteristics: no matter what purpose the user is in, he can freely run the software according to his own wishes; the user can freely learn and modify the software to help users complete their calculations as a prerequisite. The user must have access to the source code of the software; the user can freely distribute the software and its modified copies, and the user can share the improved software with the entire community for the benefit of others.

Free software is a kind of free computer software that the developer owns the copyright and reserves the right to control the distribution, modification, and sale. The source code is usually not released to prevent users from modifying the source code.

In a broad sense, free software is a subset of open source software, and the definition of free software is stricter than that of open source software. At the same time, open source software requires the source code to be attached when the software is released, and it is not necessarily free; similarly, free software is just the software provided to users for free, not necessarily open source. The relationship between open source software, free software, and free software is shown in Fig. 1.20.

The open source software market is widely used. According to a Gartner survey, 99% of organizations use open source software in their IT systems. At the same time, open source software is widely used in server operating systems, cloud computing, and the Web.

The scale of the open source software market ranks first among server operating systems. According to statistics, more than 90% of the operating systems running on global public cloud vendors' servers are open source

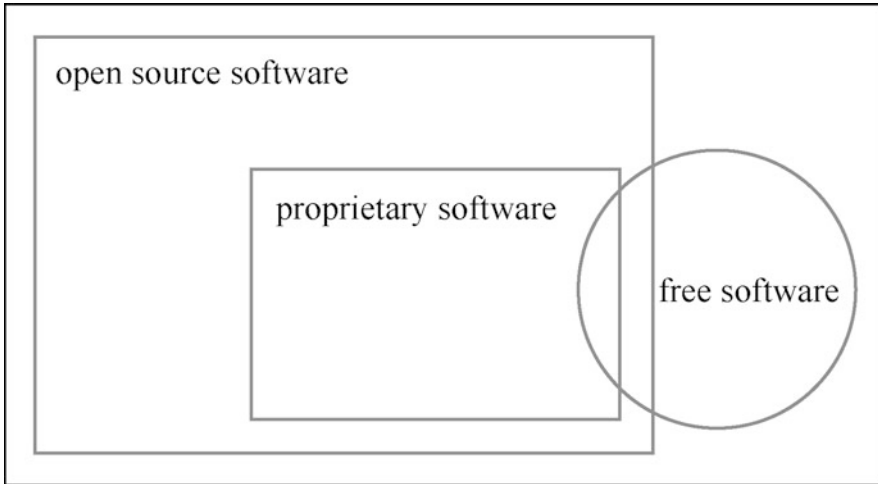


Fig. 1.20 The relationship between open source software, proprietary software and free software

Linux operating systems. Its market share in the embedded market is 62%, and its market share in the field of supercomputing is even higher. Reached 99%. The Android system based on the Linux kernel runs on more than 80% of smart phones in the world.

The 2019 annual report released by GitHub, the world's largest developer community, revealed a data: GitHub currently has more than 40 million developer users worldwide, 80% of which are from outside the United States, and the use of open source in China is proliferating. In 2018 alone, nearly 10 million new developer users joined the GitHub community and contributed to 44 million open source projects worldwide.

Open source software is also widely used in the field of cloud computing. Open source in the cloud computing field is currently mainly based on two levels: IaaS and PaaS. The IaaS level includes OpenStack, CloudStack, oVirt, ZStack, etc., and the PaaS level includes OpenShift, Rancher, Cloud Foundry, and the scheduling platform Kubernetes, Mesos, etc. For example, OpenStack software is widely used in IT, telecommunications, research, finance, and other fields. Another example is the open source application container engine Docker. Since its release in 2013, its technology has become increasingly mature. At present, the number of container image downloads has exceeded eight billion times.

The wide application of open source software in other fields also includes big data, Software Defined Network (SDN), Network Function Virtualization (NFV), artificial intelligence, and other areas. For example, big data basic analysis platforms include Hadoop, Spark, etc., NFV has OPNFV, and artificial intelligence has TensorFlow.

1.8.2.1 The Significance of Open Source

1. The open source ecology promotes national information technology innovation and drives economic development.

Open source effectively promotes technological innovation. The open source model can effectively achieve information exchange, obtain newer source codes of key technologies, use global technical resources to promote technological development iterations rapidly, break technical barriers, and promote new technologies' popularization.

Open source can realize software autonomous and controllable. Open source is more transparent and open. The establishment of my country's open source software industry can effectively guarantee information security, achieve autonomy and control, and ensure that information security is easier to manage. Products and services generally do not have malicious backdoors, and vulnerabilities can be continuously improved or patched.

Open source promotes the development of education and scientific research. Open source provides more independent learning resources for teachers and students in colleges and universities. Students can join open source projects directly and quickly, with continuous technical level improvement and continuous enrichment of experience.

Open source promotes the development of industry informatization. The open source model can effectively reduce application costs and technical thresholds, accelerate enterprise informatization, and promote the vigorous development of my country's economy.

2. Software vendors rely on open source technology to enhance R&D capabilities.

Software vendors use open source technology to reduce research and development costs, communicate through open source communities, and are familiar with the use of open source technology, which facilitates tracking of open source technology updates and adapts them according to business needs.

Well-known open source projects have the participation of high-level R&D personnel in the industry, and their source code has many references for technical personnel in terms of coding style and algorithm ideas. In using open source projects or conducting secondary development based on open source projects, software manufacturer R&D personnel can learn innovative methods to solve problems by reading source code and other methods.

After software vendors open the project, the project has a wider range of users and more complex application scenarios. R&D personnel should consider the company's business needs and personnel usage when developing but need to pay more attention to code compatibility and standardization.

3. Users use open source technology to change the route of informatization.

Enterprise users can carry out customized development based on open source technology. The functions that end-user information systems need to achieve are different. Compared with closed-source software, open source software is more flexible and has a higher degree of customization. End enterprise users can do

secondary development based on open source code to achieve the requirements of specific scenarios and specific functions and avoid binding risks.

Use open source technology to allow companies to focus on innovation. As more business resources get rid of developing software shackles, the focus of enterprises will shift to innovation. Creativity flourishes in small- and medium-sized enterprises because they are better able to create competitive alternative technologies and proprietary software to obtain more unique and forward-looking thinking than their competitors.

4. Enterprises independently open source and lead the technological development path.

Enterprises' independent open source can effectively improve R&D efficiency and enhance the quality of code. The project's open source process can attract outstanding developers and users to participate in it, inject more "fresh blood," and allow the project to continue to develop. At the same time, open source projects are deployed in different application scenarios, exposing more problems in the project and saving test costs.

Enterprises' independent open source can lead the development of technology and establish an ecosystem with open source enterprises as the core. During the operation of open source projects, potential users can be attracted to use open source software, so that more companies and developers in the industry can understand the technological development of the companies that belong to the open source project, and establish an upstream and downstream ecosystem of providers and users through open source technology, and keep abreast of users. Demand, seize the business territory, and drive the healthy development of the enterprise.

1.8.2.2 Open Source Is a Methodology

The vital impact of open source is that it makes learning programming easier. Any novice can access countless mature products for free as a reference, and the novice will one day become an experienced developer and feedback the open source community. Therefore, the open source community can develop continuously and sustainably, and the open source culture has become the programmer community's representative culture.

Open source has two aspects: one is open source software technology and related aspects, including open source software history, open source software agreements, technical products, open source communities, related hardware, technical personnel, open source software-related industries and enterprises. The second is open source values and methodology. Relevant content includes open source value system, open source methodology system, non-technical projects carried out with open source methodology, related non-technical organizations, communities, and people. Open source values' connotation mainly includes six aspects: dedication, sense of gratitude, open spirit, courage, pursuit of continuous progress, and the spirit of obtaining fair value returns based on labor. The connotation of open source methodology

mainly includes promoting progress and innovation through open sharing, solving complex and systemic problems through gathering and accumulating the labor and wisdom of many references, completing open source projects through community platforms, and through effective organization and organization of well-known companies and individuals to develop and complete the project. Open source values and open source methodology are valuable spiritual wealth contributed by open source technology to mankind.

1.8.2.3 Open Source Brings Challenges to Cloud Computing

The open source development model provides a new way for the revolution and transformation of the industry model, and the open source software resources provide directly usable software technologies, tools, and products for the development of the IT industry. The open source development model's openness and transparency help to quickly gather public wisdom and effectively promote the formation and development of technology and application ecology.

Since the development of open source, the types of software products have become more and more diverse, their functions have become more and more powerful, and the scale of the community has become larger and larger. Accordingly, the tools and technical systems supporting open source development have become more difficult to master. The new situation has brought challenges to the further prosperity and development of open source software and put forward higher requirements for open source participants. In terms of participation in open source software development, studies have pointed out that the number and proportion of long-term contributors in well-known open source communities has declined in recent years. The survey found that in the application of open source products and technologies, quite a few IT companies lack talents who master open source technologies. Only by effectively cultivating open source talents can the team of open source contributors be expanded.

Compared with the traditional software development field, open source has different or even broader requirements for software talents, so that the conventional talent training system may need to adapt to the needs and pay attention to specific aspects of open source talent education.

1.9 Exercise

(1) Multiple Choice

1. The “cloud” in “cloud computing” is more credible about its origins ().
 - A. Some aspects of cloud computing are as elusive as the cloud.
 - B. The supporting technology of cloud computing is often represented by a cloud-like pattern on the Internet, so the network that provides resources is often called “cloud.”
 - C. The scale of cloud computing is generally as broad as the cloud.
 - D. Users cannot see the resources contained in cloud computing, as if hidden behind the cloud.
2. () is not a major feature of cloud computing.
 - A. On-demand self-service.
 - B. Extensive network access.
 - C. Complementary resources.
 - D. Fast elastic scaling.
3. Systems or services that do not rely on cloud computing are ().
 - A. Baidu net disk.
 - B. Attention to the conference system.
 - C. There are Cloud Notes.
 - D. The remote login system for the supercomputer.
4. The misconception about resource pooling is ().
 - A. Resource pooling is one of the prerequisites for on-demand self-service.
 - B. Resource pooling is equivalent to resource classification.
 - C. Resource pooling requires that all resources are decomposed to a minimum unit.
 - D. Resource pooling masks the differences between different resources.
5. The misconception about the rapid elastic scaling of cloud computing is () .
 - A. Elastic expansion is considered one of the core reasons to engage users in cloud computing.
 - B. Rapid elastic scaling means that cloud users can automatically and transparently scale their IT resources according to their needs.
 - C. Rapid elastic scaling must be manually expanded or reduced.
 - D. Rapid elastic scaling enables users to save money while keeping their business or applications running smoothly.
6. The misconception about the metering service for cloud computing is ().
 - A. Metering is the basis of billing.
 - B. Services in cloud computing are measured based on the time of use.

- C. Billing management systems are commonly available in cloud computing systems and are designed to collect and process usage data.
 - D. Using a quota billing system prevents further usage requests from cloud users when the quota is exceeded.
7. () is not a service provided by cloud computing.
- A. IaaS.
 - B. PaaS.
 - C. SaaS.
 - D. RaaS.
8. () is not a key driver of the birth and development of cloud computing technology.
- A. Increase in network bandwidth B. The emergence of deep learning techniques.
 - B. The emergence of virtualization technology D. The development of the mobile Internet.
 - C. Enter the era of big data.
9. The benefits of cloud computing do not include:
- A. Cost savings.
 - B. Data is instantly accessible from anywhere.
 - C. Improve adaptability and scale IT needs flexibly. D. Enhance the security and confidentiality of user data.
10. Multi-tenant technology is an important support technology for cloud computing. () is not a general feature of multi-tenant applications.
- A. Use isolation.
 - B. B. Data security.
 - C. C. Recoverability.
 - D. Scalability E. Synergy.

(2) Fill in the Blanks

1. Cloud computing technology provides computing resources, _____, and other various resources to resource users in the form of services through the network.
2. _____ technology is the basic support of cloud computing. The cloud is inseparable from the _____ network.
3. The English abbreviation of the Internet service provider that provides network access services for cloud services is _____.
4. According to the classification of cloud computing operation mode, cloud can be divided into _____, _____, community cloud, hybrid cloud, and industry cloud.

5. _____provides highly scalable and on-demand IT capabilities based on hardware resources such as servers and storage in the form of services. Usually charged according to the cost of the resources consumed.
6. _____is located in the middle of the three-tier service of cloud computing, usually also called “cloud operating system,” which provides end-users with an Internet-based application development environment, including application programming interfaces and operating platforms, etc.

(3) Answer the Following Questions

1. What is the definition of cloud computing?
2. What are the features of cloud computing? What are the benefits of using cloud computing?
3. What are the types of cloud computing that can be divided into operational models? What kind of security and privacy guarantees?
4. What do cloud computing have in common with traditional host computing?
5. What are the key technologies included in cloud computing technology, called cloud enabling technology?
6. What are three service models for cloud computing? Which one does Amazon’s AWS and Microsoft’s Windows Azure belong to?
7. How to understand cloud computing from the perspective of triadic epistemology?
8. What are the differences and connections between open source software, free software, and free software?

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Chapter 2

Cloud Computing System



Cloud computing is based on a variety of existing technologies and is a master of existing information technologies. Cloud computing has the characteristics of being technology-centric, and its architecture includes a variety of technical mechanisms. This chapter will start with four aspects: cloud infrastructure mechanism, cloud management mechanism, cloud security mechanism, and basic cloud architecture, and discuss some of the main technical mechanisms for building the foundation of cloud technology architecture.

2.1 Cloud Infrastructure Mechanism

Just as a house has foundations, walls or columns, slabs and floors, stairs, roofs, doors, windows, and other main building components, the cloud computing environment, as a complex IT resource integrated system, also has some basic building blocks, called cloud infrastructure. These facilities are the basis for building a basic cloud environment. This section will introduce five main cloud infrastructures, namely Logical Network boundary, Virtual Server, Cloud Storage Device, Cloud Usage Monitoring, and Resource Replication.

2.1.1 Logical Network Boundary

The logical network boundary can be regarded as a relatively independent area. This area uses firewalls and other network devices to isolate a certain network environment from other network parts, forming a virtual network boundary.

The logical network boundary isolates a group of related cloud-based IT resources from other entities in the cloud (such as unauthorized users). The resources in the logical network boundary may also be physically distributed in different areas.

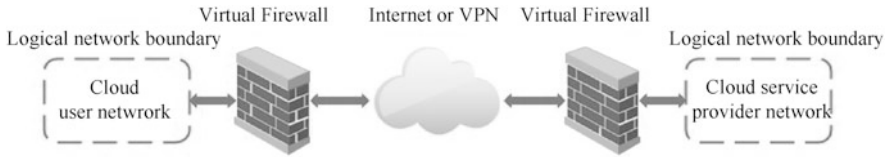


Fig. 2.1 Two logical network boundaries including cloud users and cloud service provider network environments

To provide a virtualized isolation mechanism, there must be corresponding network equipment. These network equipment usually include the following.

- **Virtual Firewall:** Can actively filter the network traffic of the isolated network and control its interaction with the Internet.
- **Virtual Network:** Generally formed by virtual local area network (VLAN), used to isolate the network environment in the data center infrastructure.

Logical network boundaries are generally deployed as virtualized IT environments.

Figure 2.1 shows two logical network boundaries, one is the internal enterprise environment containing cloud users, and the other is the cloud environment belonging to the cloud service provider. The two logical network boundaries are connected through the Internet or a virtual private network (VPN). The advantage of using VPN is that the data between the two parties can be encrypted and the communication content can be protected.

The logical network boundary's main function is network segmentation and isolation to ensure the relative independence of the IT facilities in the area. Usually, the network infrastructure needs to be virtualized before the logical network boundary forms a logical network layout. A virtual firewall based on a physical firewall is used for network isolation. For example, an enterprise may divide its entire network into an internal network and an external network. The enterprise's internal network is a virtual network isolated by the internal firewall and can only access resources within the internal network. For example, the data in the internal data center cannot access the external network and the Internet; the external network located outside the external firewall is directly connected to the Internet; the area located between the external firewall and the internal firewall is called the control zone (Demilitarized Zone) (DMZ). The DMZ is abstracted as a virtual network, which generally includes a proxy server and a Web server. The proxy server is responsible for coordinating access to common network services (DNS, E-mail, and Web), while the Web server provides webpage access functions. The company's network is divided into three logical network boundaries: extranet, intranet, and DMZ.

2.1.2 Virtual Server

A virtual server is a server that is virtualized on a physical server with virtualization software. Through virtualization, applications' monopoly of physical resources is avoided, and the same physical server can generate multiple virtual server instances. By providing independent virtual server instances to cloud users, cloud service providers can use limited physical servers and other IT resources and improve resource utilization efficiency.

A virtual server is the most basic building block of a cloud environment. Each virtual server can store a large amount of IT resources, such as CPU, memory, external storage, and network. To facilitate users to create virtual server instances, cloud service providers or users usually prepare some customized virtual server images in advance. As mentioned earlier, the several products listed in the Huawei Elastic Cloud Server ECS can actually be regarded as customized virtual server images. Each product generally specifies the number of virtual CPUs contained in its corresponding virtual server instance. Specific performance indicators such as clock speed, memory, and network bandwidth. Creating a virtual server instance from an image file is a resource allocation process that can be completed quickly and on-demand.

By installing or releasing virtual servers, cloud users can customize their own environment, which is independent of other cloud users based on virtual servers based on the same underlying physical server.

The two virtual servers in Fig. 2.2 are based on the same physical server, and they can provide services for different users.

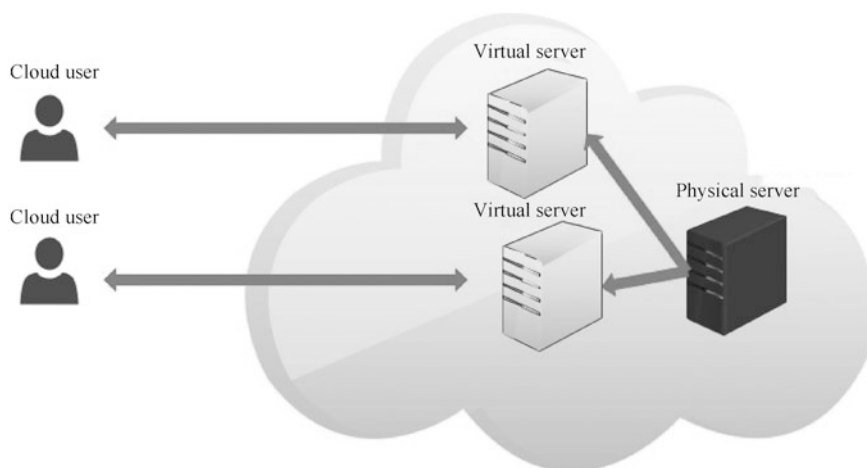


Fig. 2.2 Two virtual servers based on the same physical server can provide services to different users

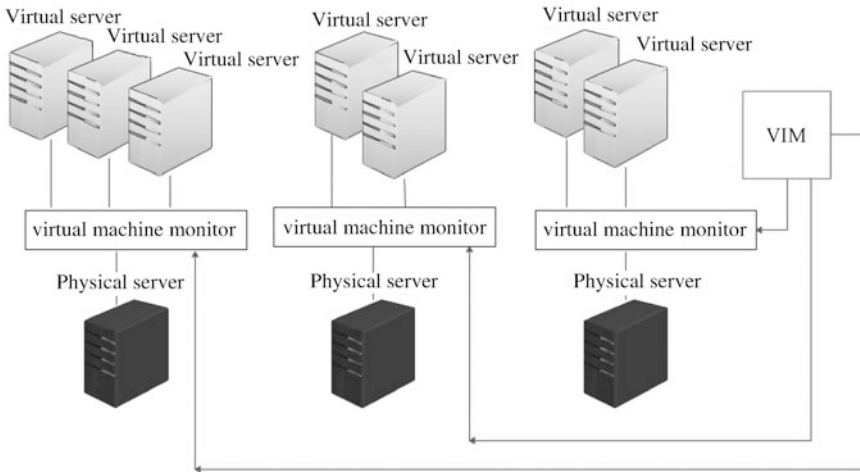


Fig. 2.3 The virtual machine monitor controls and manages virtual servers running on physical servers

Virtual Machine Monitor software is usually run on the physical server to control and manage the virtual server. A Virtual Infrastructure Manager (VIM), also known as a virtual device manager, is used to coordinate work related to the creation of virtual server instances. Figure 2.3 shows several virtual servers running on physical servers. They are created by the central VIM and controlled and managed by a virtual machine monitor.

The virtual server and virtual machine used in this book are synonymous. To quickly create a virtual machine, you can usually use a pre-customized virtual machine image (VM image).

Virtual machine images are usually stored in common cloud storage devices as files. Cloud service providers or cloud users can prepare virtual machine images with different hardware and performance specifications for users to choose according to business needs. For example, on Huawei Cloud's website, the elastic cloud server ECS provides many types of virtual machine images for different user needs, such as general computing scenarios, memory-intensive scenarios, high-performance computing scenarios, big data, and computing acceleration scenarios. The category also contains a large number of sub-categories, giving different CPU/memory ratios, virtual CPU quantity ranges, base frequency/turbo frequency, network speed, applicable scenarios, and other performance or product parameters. Cloud users can choose the most suitable product according to their needs.

2.1.3 Cloud Storage Devices

Cloud storage devices refer to storage devices that are used to provide cloud services. These physical storage devices are usually virtualized to offer services to cloud users in the form of virtual storage devices, just as physical servers are virtualized into virtual servers (virtual machines). Generally, cloud storage devices support remote access by cloud users.

Since cloud services are usually billed according to usage, cloud storage devices need to support a pay-per-use mechanism. When a cloud user and a cloud service provider negotiate a Service-Level Agreement (SLA), the user's initial and highest cloud storage capacity is usually determined. When capacity needs to be expanded, cloud storage devices can generally provide a fixed increase in capacity allocation, such as increasing 1GB each time. If the maximum capacity limit is reached, new expansion requests can be rejected, and cloud users can apply for new and larger storage capacity.

The main issues related to cloud storage are data security, integrity, and confidentiality. If the user is using a public cloud service, its data will be stored in the cloud service provider's cloud storage device. These cloud storage devices are managed and operated by cloud service providers. Cloud users have only access rights to these data, but no control rights. There is no guarantee that these data will not be peeped, tampered with, or deleted. There must be some corresponding mechanisms to ensure this data. Security. For example, the data to be saved to the cloud storage device is encrypted before storage. Or the data is divided into blocks and the order is shuffled and then stored. Or special redundant coding mechanisms such as erasure codes, regeneration codes are used to realize data storage and restore; or verify the integrity and validity of the data stored in the cloud through some special mechanisms such as data ownership verification schemes. Research in this area is also a hot spot in the field of cloud computing. If the security, integrity, and confidentiality of the data stored in the cloud cannot be guaranteed, cloud computing will lose its appeal. Especially when data is entrusted to external cloud providers and other third parties, more attention should be paid to data protection. Besides, due to the emergence of cross-regional and cross-border cloud services, legal and regulatory issues may arise when data is migrated across regions or national borders.

Cloud storage devices can be divided into four categories according to data storage levels: files (File), blocks (Block), data sets (Dataset), and objects (Object). There are three types of corresponding storage interfaces: network storage interface, object storage interface, and database storage interface.

1. Cloud storage level

The cloud storage device mechanism provides the following standard data storage levels.

(a) File

A file refers to a collection of data stored on a permanent storage device such as a hard disk. The file can be a text document, picture, program, etc.

Files usually have a file extension to indicate the file type. File storage provides external services at the file system level, and the main operation objects are files and folders. Access to data in file operations is usually sequential.

(b) Block

Block storage divides data into fixed-length data blocks, which are directly stored on permanent storage devices such as hard disks in units of data blocks. Compared with file storage, block storage does not have the concept of files and directory trees. A data block is the smallest unit of data that can be accessed independently, and its size is generally fixed. For example, the Hadoop Distributed File System (HDFS) data block length is 64 MB by default. Block storage can be regarded as the lowest storage level closest to hardware and has higher storage efficiency.

(c) Data set

Data sets generally refer to table-based data in a relational database. These data are composed of records, and each record contains fields separated by separators. These data are usually stored in a relational database, and the Structured Query Language (SQL) statement can be used to query, insert, modify, and delete the data. After entering the big data era, to improve the storage efficiency of big data, a non-relational database (NoSQL) data set storage type appeared.

(d) Object

Object storage manages data in the form of objects. The biggest difference between objects and files is that metadata (i.e., data describing data) is added to the file. In general, an object is divided into three parts: data, metadata, and object ID (object identification). Object data is usually unstructured data, such as pictures, videos, or documents. The metadata of an object refers to the related description of the object, such as the picture's size and document owner. Object ID is a globally unique identifier used to distinguish objects. Object storage generally corresponds to Web-based resources. We can use HTTP CRUD operations, that is, add (Create), read (Retrieve), update (Update), and delete (Delete) to manage stored data. Each data storage level is typically associated with some type of storage interface that corresponds not only to a specific type of cloud storage device, but also to the access protocol it uses (see Fig. 2.4).

2. Network storage interface

File storage or block storage is mainly affected by the network storage interface category, which includes storage devices that comply with industry-standard protocols, such as the Small Computer System Interface (SCSI) used for storage blocks and server message blocks. Common Internet File System (CIFS) and Network File System (NFS) for file and network storage.

The main operation objects of file storage are files and folders, and the protocols used are NFS and CIFS that comply with POSIX (Portable Operating System Interface) standards. POSIX is a set of standards developed by IEEE and ISO/IEC. Based on the existing UNIX practice and experience, the standard

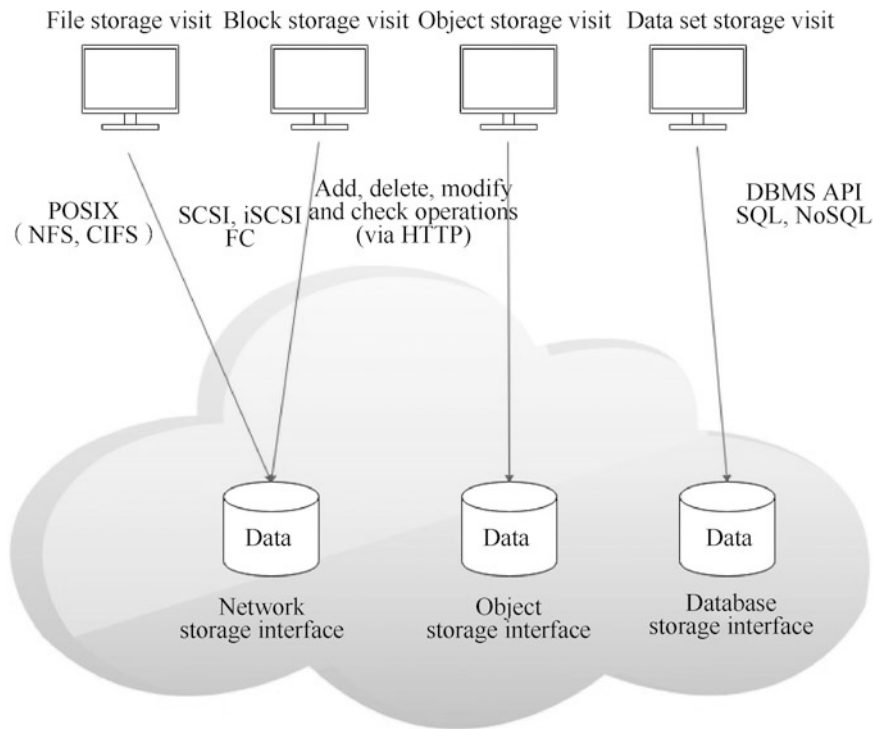


Fig. 2.4 Use different access interface technologies to operate virtualized cloud storage devices

describes the operating system's calling service interface, which ensures that the compiled application program can be transplanted and run on multiple operating systems at the source code level. Take NFS as an example. File-related interfaces include lookup (LOOKUP)/access (ACCESS)/read (READ)/write (WRITE)/create (CREATE)/delete (REMOVE)/rename (RENAME), etc. Folder-related interfaces include creating a folder (MKDIR)/deleting a folder (RMDIR)/reading a folder (REaddir), etc. Simultaneously, there are interfaces such as FSSTAT/FSINFO to provide file system-level information.

SCSI represents the protocol used by block storage. SCSI is an independent processor standard for system-level interfaces between computers and their peripheral devices (such as hard disks, optical drives, printers, scanners). The SCSI standard defines commands, communication protocols, and electrical characteristics of entities and is mainly used in storage devices (such as hard disks and tape drives). The main interfaces of SCSI are READ/WRITE/READ CAPACITY/INQUIRY, etc. FC (Fibre Channel) and iSCSI (Internet Small Computer System Interface) also block storage protocols. Compared with file storage, block storage does not have the concept of files and directory trees and usually does not define disk creation and deletion operations but pays more attention to transmission control efficiency.

The data search and extraction performance of file storage are usually not optimal. Block storage requires data to have a fixed format (called data block). This format is closest to hardware and is the smallest unit of storage and access. Compared with file storage, block storage usually has better performance.

3. Object storage interface

Various types of data can be referenced and stored as Web resources, which is object storage. It is based on technologies that support multiple data and media types. The cloud storage device mechanism that implements this interface can usually be accessed through REST with HTTP as the main protocol or cloud services based on Web services.

The main operation object of object storage is the object. Take Amazon's S3 storage as an example. The main interfaces are upload (PUT)/download (GET)/delete (DELETE), etc. Object storage does not have a random read/write interface, and there is no concept of a directory tree. The protocols it uses such as HTTP is more focused on simplicity and efficiency.

4. Database storage interface

In addition to basic storage operations, cloud storage device mechanisms based on database storage interfaces usually support query languages and implement storage management through standard APIs or management user interfaces.

According to the storage structure, there are two types of this storage interface.

(a) Relational database

Relational databases use tables to organize related data into rows and columns. Usually, it uses industry-standard SQL to operate the database. Most commercial database products use relational database management systems. Therefore, the cloud storage device mechanism implemented by using relational data storage can be based on many existing commercial database products.

The challenges of cloud-based relational databases mainly come from expansion and performance. It is difficult to scale a relational cloud storage device horizontally, and vertical expansion is more complicated than horizontal expansion. When remotely accessed by cloud services, complex relational databases and databases containing a large amount of data will have higher processing costs and longer delays.

For large databases, its performance is also closely related to the location of data storage. Local data storage is better than storage on a Wide Area Network (WAN) in terms of network reliability and latency.

(b) Non-relational database

Traditionally, data set storage corresponds to relational databases. After entering the era of big data, because relational databases are not easy to expand horizontally and their storage efficiency is low, non-relational databases are also widely used. These non-relational databases are often referred to as NoSQL. One of the main features of NoSQL databases is the removal of the relational characteristics of relational databases. Compared with traditional relational databases, a "more loose" structure is used to store data, and

no emphasis is placed on defining relationships and realizing data standardization. The database has a simple structure and is very easy to expand. Generally, NoSQL databases have very high read/write performance, especially under large amounts of data.

The main motivation for using non-relational storage is to avoid the possible complexity and processing costs of relational databases. At the same time, compared with relational storage, non-relational storage can be scaled more horizontally.

Non-relational storage generally does not support the functions of relational databases, such as transactions and connections. The storage is usually non-standard data, which limits the portability of the data.

2.1.4 Cloud Usage Monitoring

According to usage, the feature of cloud service billing makes the cloud computing system have a mechanism for monitoring and measuring cloud users' resource usage. Cloud usage monitoring is a lightweight autonomous software program used to collect and process cloud users' usage data of IT resources.

Depending on the type of usage indicators that need to be collected and the method of collecting usage data, cloud usage monitoring can exist in different forms. The following are three common agent-based forms, each sending the collected usage data to the log database for subsequent processing and reporting.

1. Monitoring agent

The monitoring agent is an intermediate event-driven program, which resides on the existing communication path as a service agent, and transparently monitors and analyzes the data flow. This type of cloud usage monitoring is usually used to measure network traffic and message metrics (see Fig. 2.5).

In this example, the cloud user sends a request message to the cloud service and the monitoring agent intercepts this message. On the one hand, the user's request message is sent to the cloud service, on the other hand, the relevant usage data is collected and stored in the log database. After the cloud service receives the request message, it will return a response message, which the monitoring agent will not intercept.

2. Resource agent

Resource agent is a processing module that collects usage data by interacting with specialized resource software in an event-driven manner. It works based on resource software, monitoring the usage indicators of predefined and observable events, such as the start, pause, resume, and vertical expansion of each entity. An example of a resource agent is shown in Fig. 2.6.

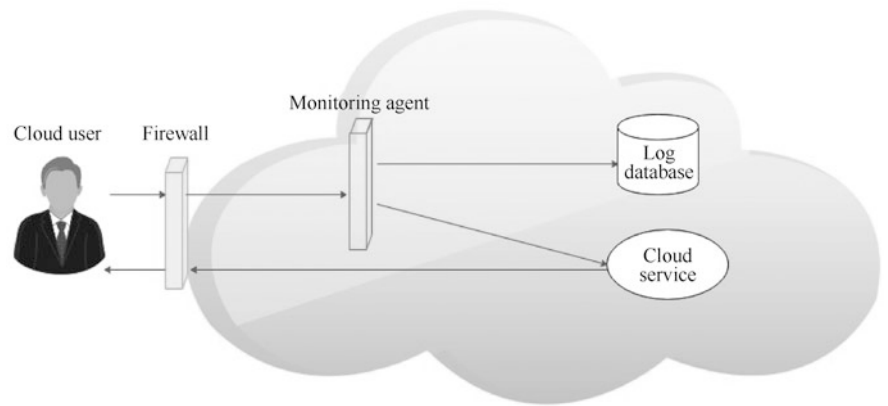


Fig. 2.5 Example of monitoring agent

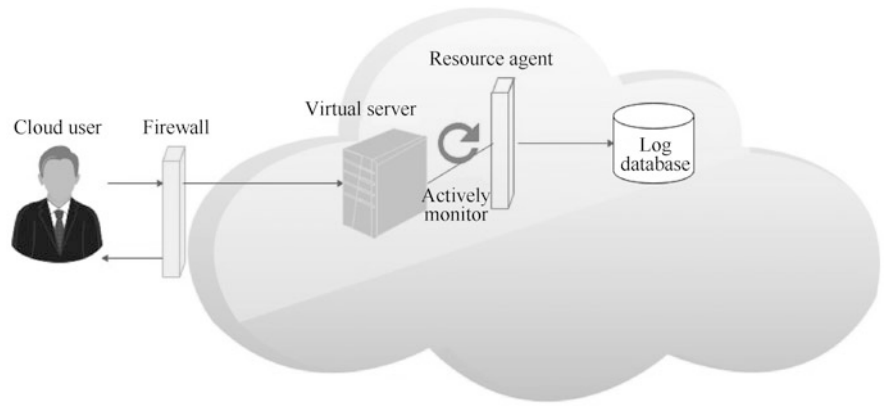


Fig. 2.6 Example of resource agent

In this example, the resource agent actively monitors the virtual server and detects increased resource usage. Specifically, the resource agent receives notifications from the underlying resource management program, and as the amount of user requests increases, the virtual server expands. The resource agent will store the collected usage data in the log database according to its monitoring indicators.

3. Polling agent

A polling agent is a processing module that collects cloud server usage data by polling IT resources. It is usually used to periodically monitor the status of IT resources, such as uptime and downtime.

The polling agent in Fig. 2.7 monitors the status of cloud services on the virtual server and sends polling messages periodically. When the received polling response changes (e.g., when the usage status changes from A to B), record the new usage status to the log database.

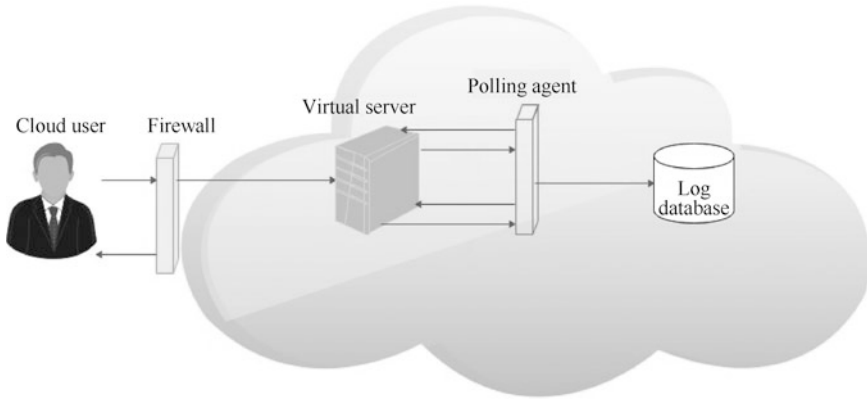


Fig. 2.7 Example of polling agent

2.1.5 Resource Replication

The resource replication mentioned here refers to using customized resource templates (such as virtual machine images) to create multiple virtual server instances. This is usually performed when the availability and performance of IT resources need to be enhanced.

The resource replication mechanism uses virtualization technology to realize the replication of cloud-based IT resources. In Fig. 2.8, the virtual machine monitor uses the stored virtual server image to replicate multiple virtual server instances.

2.2 Cloud Management Mechanism

Cloud-based IT resources need to be established, configured, maintained, and monitored. This section mainly introduces the system that contains these mechanisms and can accomplish these management tasks. They promote the control and evolution of IT resources that form cloud platforms and solutions, thus forming a key part of cloud technology architecture.

The management mechanisms or systems introduced in this section mainly fall into four categories: Remote Administration, Resource Management System, SLA Management System, and Billing Management System. These systems usually provide integrated APIs and can be provided to users in the form of individual products, customized applications, various combined product packages, or multi-functional applications.

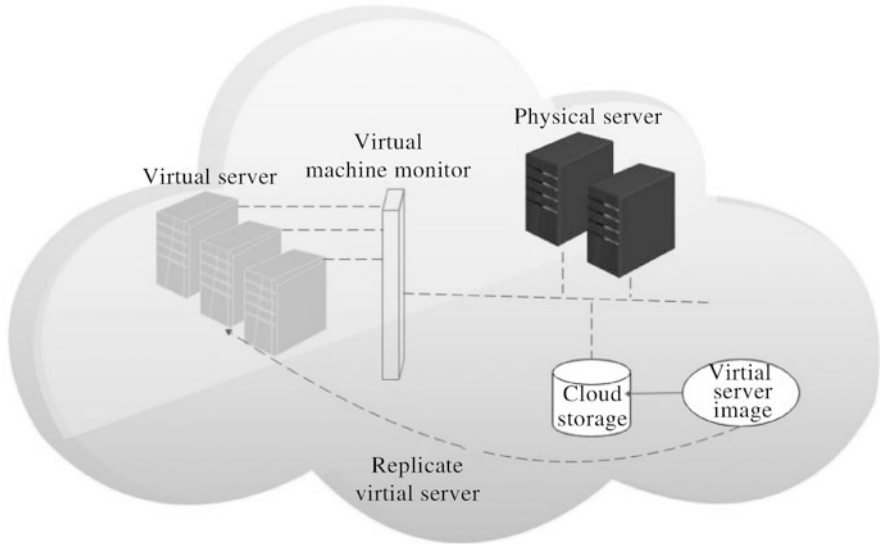


Fig. 2.8 The virtual machine monitor uses the stored virtual server image to replicate multiple virtual server instances

2.2.1 Remote Management System

The remote management system provides external cloud resource managers with tools and user interfaces to configure and manage cloud-based IT resources.

The remote management system will establish an entrance to access the control and management functions of various underlying systems, including resource management systems, SLA management systems, and billing management systems (see Fig. 2.9).

Cloud providers generally use the tools and APIs provided by the remote management system to develop and customize online portals, which provide cloud users with various management and control.

The remote management system mainly creates the following two types of entrances.

1. Usage and administration portal

A universal portal that centrally manages different cloud-based IT resources and provides IT resource usage reports. This portal is an integral part of many cloud technology architectures.

2. Self-service portal

It is a portal that allows cloud users to search for the latest cloud services and IT resources (usually available for cloud users to rent) lists provided by cloud service providers. Then, cloud users submit their options to cloud service providers for resource allocation.

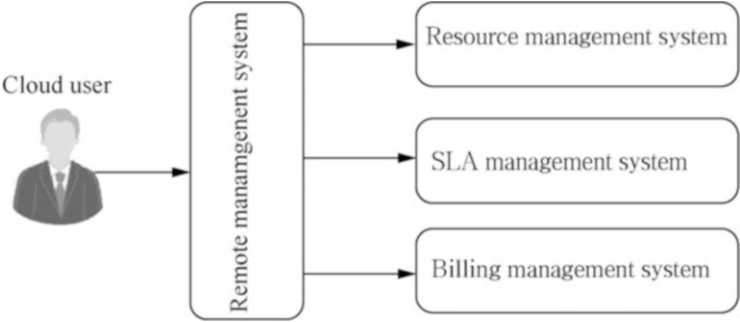


Fig. 2.9 Centralized management and control of external cloud resource managers through the remote management system

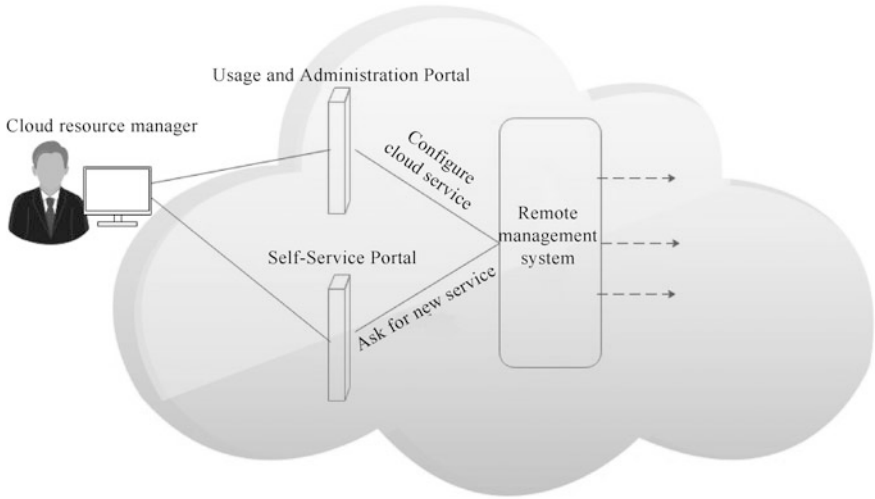


Fig. 2.10 Cloud resource managers use two types of portals

Figure 2.10 shows an example of a cloud resource manager using two portals to access a remote management system. Among them, the cloud resource manager requests the cloud service provider to provide a new cloud service through the self-service portal; after the cloud service provider provides the service, the cloud resource manager completes the configuration of the newly provided cloud service through the use and management portal.

Through the remote management console, cloud users (cloud resource managers) can perform the following tasks.

- Configure and establish cloud services.
- Provide and release IT resources for on-demand cloud services.

- Monitor the status, usage, and performance of cloud services.
- Monitor the implementation of QoS and SLA.
- Manage rental costs and usage expenses.
- Manage user accounts, security credentials, authorization, and access control.
- Track internal and external access to rental services.
- Plan and evaluate the supply of IT resources.
- Capacity planning.

2.2.2 Resource Management System

The resource management system helps coordinate IT resources in order to respond to management operations performed by cloud users and cloud service providers. Its core is VIM, which is used to coordinate server hardware so that virtual server instances can be created based on the most suitable underlying physical server. VIM is a commercial product used to manage a series of IT resources across multiple physical servers.

The tasks that are usually automated and realized by the resource management system are as follows:

- Manage virtual IT resource templates, such as virtual server images.
- Allocate, release, and coordinate virtual IT resources in the available physical infrastructure.
- Monitor the operating conditions of IT resources and enforce usage policies and security regulations.

Figure 2.11 shows two different access methods for resource management. Among them, the cloud resource manager of the cloud user accesses the use and management portal from the outside (① in the figure), and the cloud resource manager of the cloud service provider uses the local user interface provided by VIM to perform internal resource management tasks (② in the figure).

2.2.3 SLA Management System

SLA is a contract between a network service provider and a customer, which defines service type, service quality, and customer payment.

The SLA management system represents a series of commercially available cloud management products, and its functions include the management, collection, storage, reporting, and runtime notification of SLA data.

When deploying an SLA management system, it often includes a library (QoS measurement library) for storing and retrieving collected SLA data based on predefined indicators and report parameters. Collecting SLA data also requires one or more SLA monitoring agents. Through the monitoring agents, cloud resource

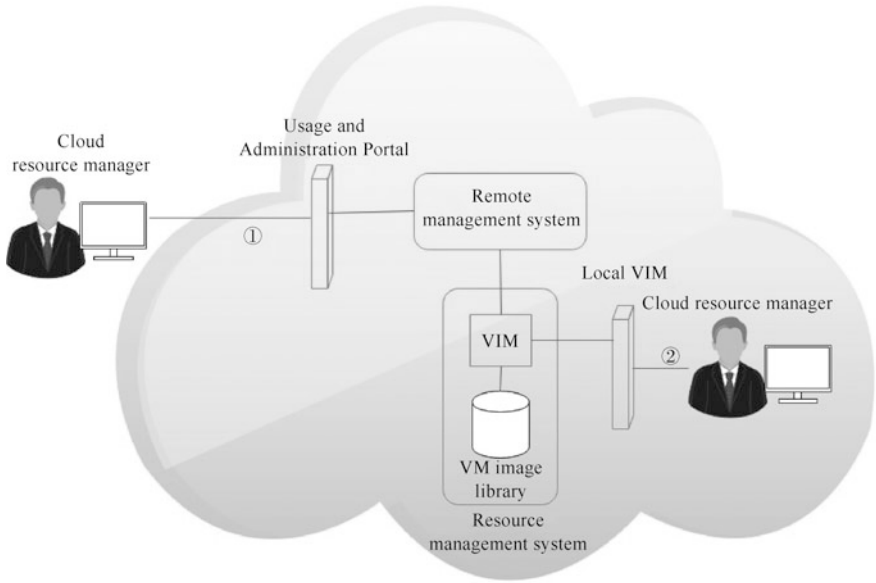


Fig. 2.11 Two different access methods for resource management

managers can query and obtain these data in approximately real-time through the use and management portal.

Figure 2.12 shows the SLA interaction between cloud users and cloud services. First, the cloud user sends an interaction message to the cloud service, the SLA monitoring agent intercepts the interaction message, evaluates the interaction and collects relevant runtime data, which is related to the service quality assurance defined in the cloud service SLA; then, the SLA monitoring agent will The collected data is stored in the QoS measurement library, which is also part of the SLA management system; finally, external cloud resource managers can issue queries and generate reports through the use and management portal, and internal cloud resource managers can directly query SLA management system.

2.2.4 Billing Management System

The billing management system is specifically used to collect and process the usage data of cloud users. It involves the settlement of cloud service providers and the billing of cloud users. Specifically, the billing management system relies on pay-per-use monitors to collect user usage data during operation. These data are stored in a library (pay-per-use measuring library) inside the system.

Figure 2.13 shows the interaction between cloud users and cloud services related to billing. Among them, when a cloud user accesses a cloud service, a pay-per-use

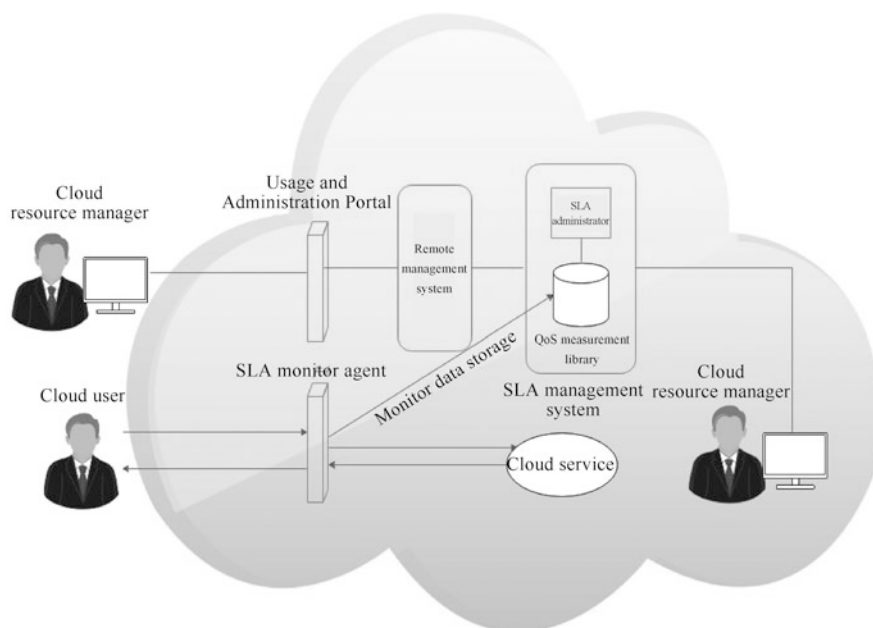


Fig. 2.12 SLA-related interaction between cloud users and cloud services

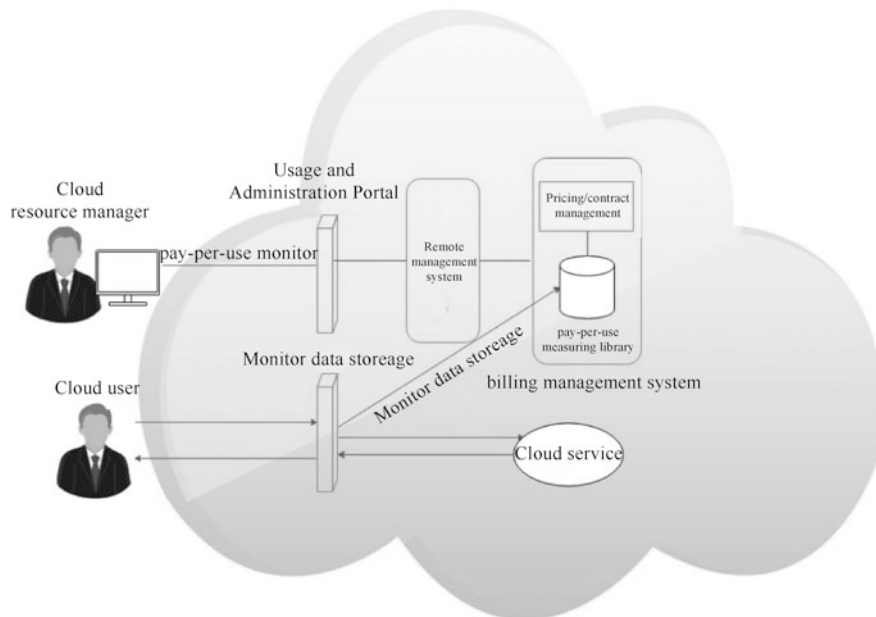


Fig. 2.13 Billing interaction between cloud users and cloud services

monitor tracks usage, collects billing-related data, and sends it to the library in the billing management system. The system regularly calculates usage fees and generates invoices for users. Invoices can be provided to cloud users or cloud resource managers through the use and management portal.

2.3 Cloud Security Mechanism

Users move their business data into the cloud, which means that cloud service providers have to bear the responsibility for this part of data security. The remote use of IT resources requires the cloud user to extend the trust boundary to the external cloud used by the user. It is very difficult to establish a security architecture that includes such a trust boundary without introducing security vulnerabilities. This section will introduce a set of basic cloud security mechanisms to counter the security threats that may be encountered in cloud services.

2.3.1 Encryption

An encryption mechanism is a digital coding system used to protect data to ensure its confidentiality and integrity. It is used to convert Plaintext into a protected, unreadable format. The Plaintext here refers to data encoded in a readable format.

Encryption technology usually relies on a standardized algorithm called an encryption component (Cipher) to convert the original Plaintext into ciphertext, encrypted data. If you don't know the secret key, even if you get the ciphertext, unless you master superb deciphering techniques, you generally cannot analyze the original Plaintext from it, except metadata such as the length of the message and the creation date.

When encrypting the Plaintext, a key is needed. The key is a secret message established and shared by authorized parties. The key is also needed to decrypt the ciphertext.

Encryption mechanisms can help combat security threats such as traffic eavesdropping, malicious media, insufficient credit, and overlapping trust boundaries.

There are two common encryption keys: Symmetric Encryption and Asymmetric Encryption.

Figure 2.14 shows an example of using encryption mechanisms to defend against malicious service agent attacks. After the document is encrypted, the malicious service agent cannot obtain data from the encrypted message.

1. Symmetric encryption

Symmetric encryption uses the same key for encryption and decryption. The encryption and decryption processes are all done by authorized parties using a

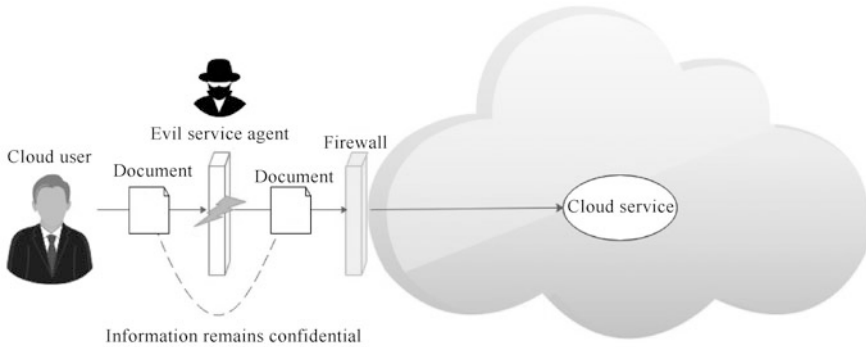


Fig. 2.14 After encryption, the content of the document remains confidential through the untrusted transmission channel

shared key, that is, a message encrypted with a specific key can only be decrypted with the same key.

Symmetric encryption has no Non-Repudiation. If more than one party owns the key, it is impossible to determine which party performed the message encryption or decryption.

2. Asymmetric encryption

Asymmetric encryption uses two keys, Public Key and Private Key. In asymmetric encryption, only the owner has the private key, and the public key is generally publicly available. A document encrypted with a certain private key can only be decrypted correctly with the corresponding public key. Similarly, a document encrypted with a certain public key can only be decrypted with the corresponding private key.

Because two different keys are used, asymmetric encryption is generally slow to calculate, but encryption with a private key can provide authenticity, non-repudiation, and integrity protection. The private key is only owned by the user. If a document is encrypted with the user's private key, he cannot deny that the document was not sent by himself. The document can only be decrypted with the user's public key, and a third party cannot tamper with its content. Similarly, the message encrypted with the public key can only be decrypted by the legal owner of the private key, and the third party cannot snoop on its content, which provides confidentiality protection for the data.

2.3.2 Hashing

When a one-way, irreversible form of data protection is needed, a hashing mechanism is used. Hashing is a commonly used technique in mathematics. It can also be translated as "hash" or transliterated as "hash." It transforms an input of any length

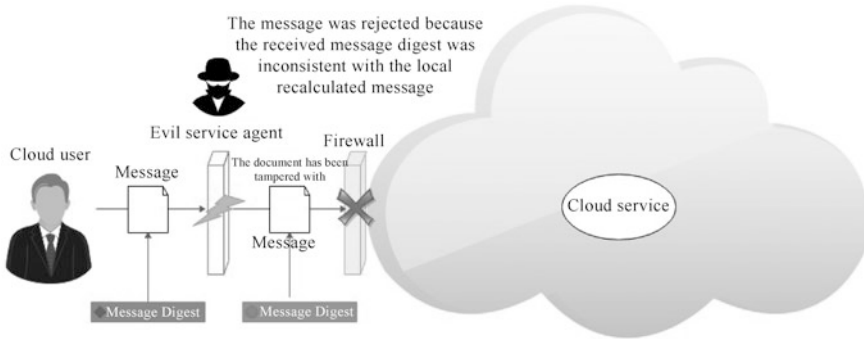


Fig. 2.15 Hashing mechanism's protection of message integrity

into a fixed-length output through a hashing algorithm. The output is the hash value. This conversion is a compression mapping, that is, the space of the hash value is usually much smaller than the space of the input, and different inputs may be hashed into the same output, so the unique input cannot be determined based on the hash value.

But conversely, if the hash values are different, it can be concluded that the inputs are also different. So hashing is simply a function that compresses messages of any length into a fixed-length Message Digest. If you compare a message to a person, and hash the message, it's like taking a low-pixel photo of the person, which can roughly depict the person's appearance and outline, as long as the compared photos are exactly the same. If the photos are taken under different conditions, it can be concluded that the people being photographed are also different.

A common application of the hashing mechanism is the storage of passwords.

Hashing techniques can be used to obtain the message digest of the message. The message digest is usually fixed in length and smaller than the original message size. The message sender can append the message digest to the message, and the receiver uses the same hash function to regenerate the message digest on the received message and compare it with the original message digest attached to the message. Determine whether the original message has been tampered with by observing whether they are consistent. Using a suitable hash algorithm, under the same computing conditions, the probability of two different messages generating the same message digest is very small. So in general, as long as the original message is modified, a completely different message digest will be generated. Therefore, as long as the newly generated message digest is the same as the original message digest sent with the message, it can be concluded that the message has not been tampered with during transmission. This method can ensure the integrity and consistency of the message.

Figure 2.15 shows the protection of message integrity by the hashing mechanism. In the figure, a cloud user wants to transmit a document to the cloud service, and the document is summarized by a hash function in advance and attached to the transmitted document. If the document is intercepted and tampered with by a malicious

service agent, the firewall will recalculate the summary of the received document and compare it with the original summary before the document enters the cloud service. If it is inconsistent, it proves that the document has been tampered with and the document is rejected.

2.3.3 Digital Signature

A digital signature is a string of information that can only be generated by the sender of the information that others cannot forge. This string of information is also an effective proof of the authenticity of the information sent by the sender. It is a kind of ordinary physical signature similar to that written on paper, but it is realized by the technology in the field of public key encryption. A set of digital signatures usually defines two complementary operations, one for signing and the other for verification. Digital signature is the application of asymmetric key encryption technology and digital digest technology.

The digital signature mechanism is a means to provide data authenticity and integrity through identity verification and non-repudiation. Before sending the message, give the message a digital signature. If the message is subsequently modified without authorization, the digital signature will become illegal. The digital signature provides a proof that the received message is consistent with the one created by the legitimate sender.

The creation of digital signatures involves hashing and asymmetric encryption. It is actually a message digest encrypted by the private key and appended to the original message. The receiver needs to verify the legality of the digital signature, decrypt the digital signature with the corresponding public key, and get the message digest. At the same time, a hash mechanism is applied to the original message to get the message digest. Two different treatments give the same result, which indicates that the message maintains its integrity.

Digital signature mechanisms can help counter security threats such as malicious media, insufficient authorization, and overlapping trust boundaries.

The malicious service agent in Fig. 2.16 intercepted and tampered with a legitimate cloud user's data-signed document and pretended to be the cloud user to request cloud services, but the digital signature of the document was verified as invalid when the document passed through the firewall, so the service request was rejected.

2.3.4 Public Key Infrastructure

Public Key Infrastructure (PKI) is a system composed of protocols, data formats, rules, and implementations. It is used to manage asymmetric key issuance so that large-scale systems can safely use public-key cryptography. This system links the

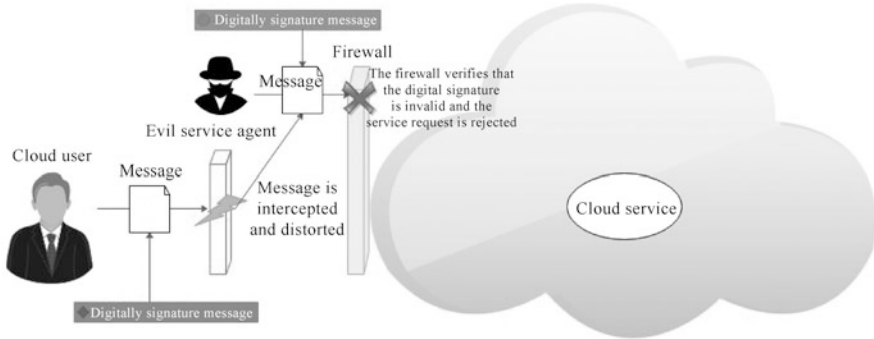


Fig. 2.16 Verify the legitimacy of the document through data signature

public key with the corresponding key owner, and at the same time, verifies the validity of the key.

PKI relies on digital certificates, which are data structures with digital signatures, which are generally issued by a third-party certificate authority (CA). The difference between digital certificates and ordinary digital signatures is that digital certificates usually carry these pieces of information: the identity information of the certificate owner, which the CA has verified; the public key of the certificate owner; the certificate issued by the CA and the digital signature of the CA. The digital signature uses the CA's private key to encrypt the digest of the certificate; there are other related information such as the validity period. A CA is generally an authority trusted by the outside world, and its public key is usually public. The outside world can use the CA's public key to verify the CA's digital signature's authenticity. If it is true, it means that the CA has verified the authenticity of the certificate owner's identity. Indirectly verified the authenticity and validity of the certificate. The typical steps for CA to generate a certificate are shown in Fig. 2.17.

Although most digital certificates are issued by a few trusted CAs such as VeriSign (manufacturers in the domain and Internet security field), digital certificates can also be generated by other methods. Larger companies such as Microsoft can act as their own CA, issuing digital certificates to other customers and the public, and even individual users can generate digital certificates as long as they have the appropriate software tools.

PKI is a reliable method for realizing asymmetric encryption, managing the identity information of cloud users and cloud service providers, and defending against malicious intermediaries and insufficient authorization threats.

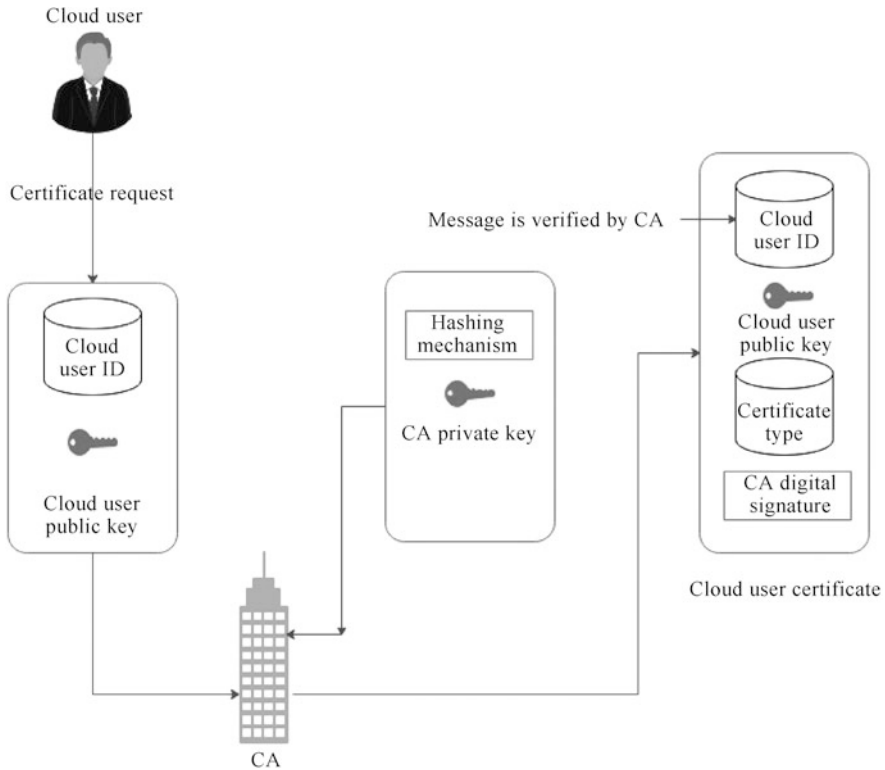


Fig. 2.17 Typical steps for certificate authorities to generate a certificate

2.3.5 Identity and Access Management

The identity and access management (IAM) mechanism includes authentication and the necessary components and strategies for managing user identities, as well as related IT resources, environment, and system access privileges.

The IAM mechanism has the following four main components.

1. Certification

In addition to supporting the most common username and password combination authentication methods, it also supports digital signatures, digital certificates, biometric hardware (such as fingerprint readers), special software (such as voice analysis programs), and the combination of user accounts and registered IP or The MAC address is bound.

2. Authorization

The authorization component is used to define the correct access control granularity and manage the relationship between user identity, access control authority, and IT resource availability.

3. User management

User management is responsible for creating new user identities and access groups, resetting passwords, defining password policies and managing privileges.

4. Certificate management

The Credential Management system has established a set of rules to manage the defined user accounts, including user identity management and corresponding access control and policies.

In addition to assigning specific user privilege levels, the IAM mechanism also includes formulating corresponding access control and management strategies. This mechanism is mainly used to combat security threats such as insufficient authorization, denial of service attacks, and overlapping trust boundaries.

2.3.6 *Single Sign On*

Cloud users sometimes need to use multiple cloud services simultaneously or continuously. If every cloud service has to re-authenticate the user, the user will be annoying. However, it is not easy to provide cloud users with authentication and authorization information across multiple cloud services. The Single Sign On (SSO) mechanism enables a cloud user to be authenticated by a security agent. This security agent establishes a security context. When the cloud user accesses other cloud services or cloud-based IT resources, the security context will be persisted. Otherwise, the cloud user has to re-authenticate himself when sending each subsequent request.

The SSO mechanism allows independent cloud services and IT resources to generate and circulate runtime authentication and authorization certificates. The certificate is first provided by the cloud user and remains valid during the session, while its security context information is shared. The concept of security context mentioned here has a wide range. It generally refers to a collection of permissions and permissions that define what an entity is allowed to do. Permissions, privileges, access tokens, integrity levels, etc. are all included in it. When cloud users want to access cloud services located in different clouds, the SSO mechanism's security agent is particularly useful.

Figure 2.18 shows an example of achieving access across multiple cloud services through single sign-on. First, the cloud user provides the security certificate for login to the security agent on the firewall. After successful authentication, the security agent responds with a security token representing the completion of the authentication. The token contains the user's identity information and can be authenticated by multiple cloud services.

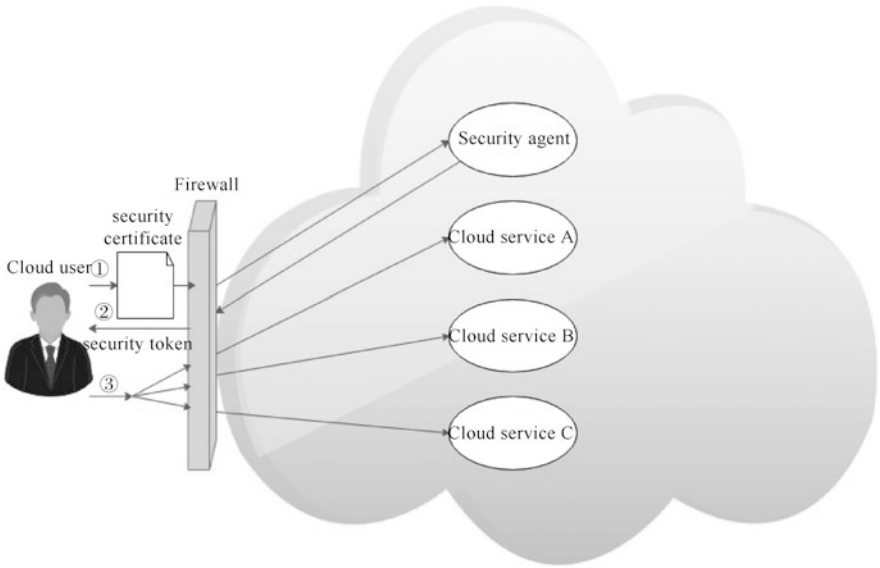


Fig. 2.18 Achieving access across multiple cloud services through single sign-on

2.3.7 Cloud-Based Security Group

Setting up isolation between IT resources can increase data protection. The separation of cloud resources creates separate physical and virtual IT environments for different users and groups, forming independent logical network boundaries. For example, according to different network security requirements, an enterprise’s network can be divided into extranets and intranets. The external network deploys a flexible firewall for external Internet access; while the internal network can only access the internal data center’s resources but cannot access the Internet.

The cloud-based resource segmentation process creates a Cloud-Based Security Group mechanism, and security policies determine the division of security groups. According to the established security strategy, the entire cloud environment is divided into several logical cloud-based security groups, and each security group forms an independent logical network boundary. Every cloud-based IT resource belongs to at least one logical cloud-based security group. The communication between these security groups is carried out through some special rules.

Multiple virtual servers running on the same physical server can belong to different cloud-based security groups.

Figure 2.19 shows an example of enhancing data protection by dividing cloud-based security groups. Among them, the cloud-based security group A includes virtual servers A and D, which are assigned to cloud user A; the cloud-based security group B includes virtual servers B, C, and E, and is assigned to cloud user B. Even if cloud user A’s certificate is compromised, the attacker can only attack the virtual

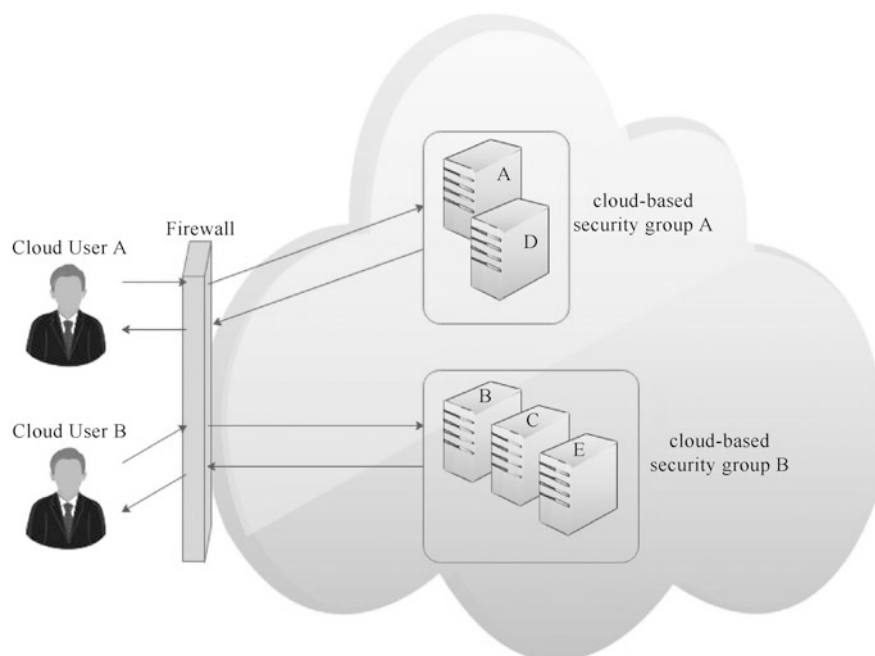


Fig. 2.19 Enhance data protection by dividing cloud-based security groups

servers A and D in the cloud-based security group A. The virtual servers B, C, and E in the cloud-based security group B cannot be affected.

2.3.8 Hardened Virtual Server Image

A virtual server is created from a template called a virtual server image (or virtual machine image). Hardening is the process of stripping unnecessary software from the system and limiting potential vulnerabilities that attackers may exploit. Removing redundant programs, closing unnecessary server ports, closing unused services, internal root accounts, and guest access permissions are all examples of enhancements.

Hardened Virtual Server Image is a hardened template used to create virtual server instances, which is usually safer than the original standard image.

Enhanced virtual server images can help combat security threats such as denial of service, insufficient authorization, and overlapping trust boundaries.

Figure 2.20 shows an example of an enhanced virtual server image. Among them, the cloud service provider applies its security strategy to the enhanced virtual server image. As part of the resource management system, the enhanced virtual server image template is stored in the VM image library.

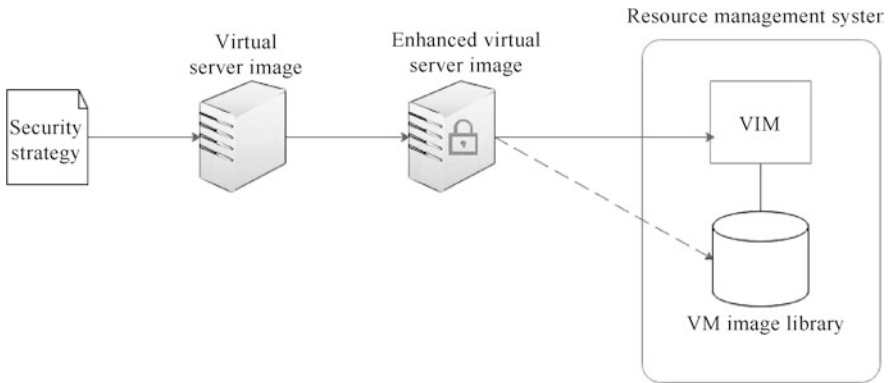


Fig. 2.20 Enhanced virtual server image

2.4 Basic Cloud Architecture

This section will describe some common basic cloud architectures. These architectures are common in modern cloud environments and are usually an important part of cloud environments.

2.4.1 Load Distribution Architecture

The level of IT resources can be expanded by adding one or more IT resources of the same kind, and the load balancer that provides runtime logic can evenly distribute the workload on the available IT resources. The resulting Workload Distribution Architecture relies to a certain extent on complex load balancing algorithms and runtime logic to reduce overuse or underuse of IT resources.

The load distribution architecture can be used to support distributed virtual servers, cloud storage devices, and cloud services. In fact, this basic architecture can be applied to any IT resource.

Figure 2.21 shows an example of a load distribution architecture. Among them, cloud service A has a redundant copy on virtual server B. If the load on cloud service A is too large, the load balancer will intercept cloud user requests and locate them on virtual servers A and B, respectively, to ensure uniform load distribution.

In addition to the basic load balancing mechanism and the virtual server and cloud storage device mechanism that can be used for load balancing, the following mechanisms are also part of the cloud architecture:

- Audit monitor.
- Cloud usage monitor.
- Virtual machine monitor.
- Logical network boundary.

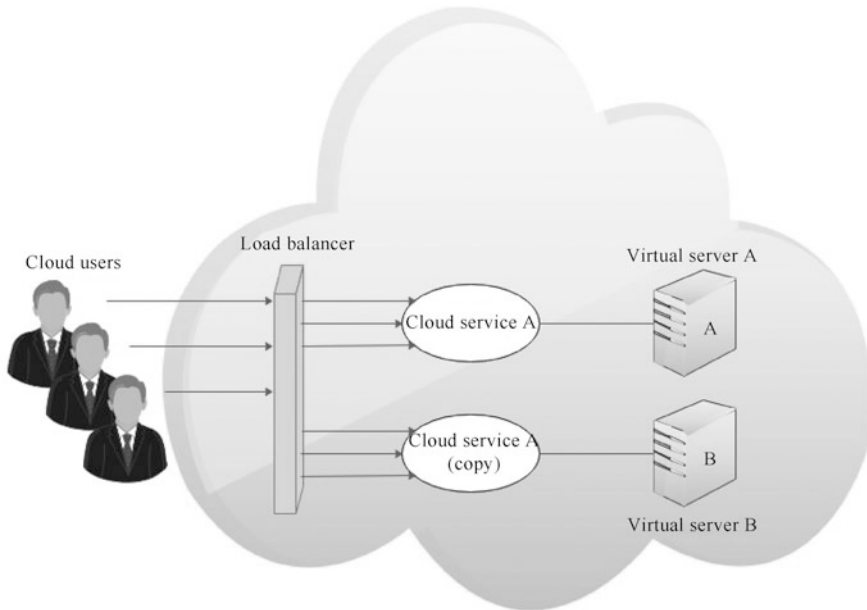


Fig. 2.21 Example of load distribution architecture

- Resource cluster.
- Resource replication.

2.4.2 Resource Pooling Architecture

Resource Pooling Architecture is based on the use of one or more resource pools, where the same IT resources are grouped and maintained by the same system to automatically ensure that they are kept in sync.

Common resource pools are as follows:

1. Physical server pool

The physical server pool consists of networked servers. These servers have already installed the operating system and other necessary programs and application software and can be put into use immediately.

2. Virtual server pool

The virtual server pool is generally configured through a customized available template, which is selected by cloud users from a variety of available templates during preparation. A cloud user can configure a low-end server pool, each virtual server is equipped with 2 vCPUs and 4GB of memory; it can also configure a

high-end server pool, each virtual server is equipped with 16 vCPUs and 32GB of memory.

3. Storage pool

Storage pools or cloud storage device pools generally consist of file-based or block-based storage structures.

4. Network pool

The network pool is composed of different pre-configured network interconnection devices. For example, for redundant connections, load balancing, or link aggregation, you can create a virtual firewall device pool or a physical network switch pool.

5. CPU pool

The CPU pool can be allocated to virtual servers, usually with a single virtual processing core (vCPU) as the basic unit.

6. Memory pool

The memory pool can be used as a new supply or vertical expansion of the physical server.

A dedicated pool can be created for each type of IT resource, or multiple pools of different types can be aggregated into a larger mixed pool. In this mixed pool, each individual pool is called a sub-resource pool.

2.4.3 Dynamic Scalability Architecture

Dynamic Scalability Architecture has some predefined expansion conditions model. Triggering these conditions will cause the system to allocate IT resources from the resource pool automatically dynamically. The dynamic allocation mechanism allows the number of resources available to users to change in accordance with changes in user needs.

Common types of dynamic expansion are as follows:

1. Dynamic horizontal scaling

Expand IT resources by adding or subtracting instances of the same type of IT resources to handle changes in workload. If resources need to be added, the automatic extension listener will request resource replication according to requirements and permissions and send a signal to start IT resource replication.

2. Dynamic vertical scaling

When it is necessary to adjust the processing capacity of a single IT resource, expand the IT resource instance up (enhance the configuration) or down (down the configuration). For example, when a virtual server is overloaded, its memory capacity can be dynamically increased, or a processing core can be added.

3. Dynamic relocation

Relocate service requirements to other IT resources that can provide similar services at runtime. For example, the virtual server corresponding to the cloud service is migrated to a more powerful physical host.

The dynamic expansion architecture can be applied to a range of IT resources, including virtual servers and cloud storage devices. In addition to the core automatic extension listener and resource replication mechanism, the following mechanisms are also used in this form of cloud architecture.

1. Cloud usage monitor

In response to the dynamic changes caused by this architecture, a special cloud usage monitor can be used to track runtime usage.

2. Virtual machine monitor

The dynamically scalable system calls the virtual machine monitor to create or remove virtual machine server instances or extend itself.

3. Pay-per-use monitor

The pay-per-use monitor collects usage cost information in response to the expansion of IT resources.

2.4.4 Elastic Resource Capacity Architecture

The Elastic Resource Capacity Architecture is mainly related to the dynamic supply of virtual servers. The architecture allocates and recycles related IT resources based on real-time changes in user needs, relying on the resource pool containing IT resources such as CPU and memory in the system, so as to respond to changes in user needs instantly.

The flexible resource capacity architecture monitors the needs of cloud users through the automatic extension of the monitor. Once the requirements change, it will execute the pre-deployed intelligent automation engine script, interact with the virtual machine monitor and VIM, and automatically process user requests and notify the resource pool to allocate or reclaim corresponding resources on the processing results. The elastic resource capacity architecture monitors the runtime processing of the virtual server. Before the cloud service capacity reaches the capacity threshold, it can obtain additional processing power from the resource pool through dynamic allocation. Under this architecture, virtual servers and their hosted applications and IT resources can be seen as vertically expanding.

Figure 2.22 shows an example of a flexible resource capacity architecture. The cloud user actively sends a request to the cloud service, and the automatic extension listener monitors this. The intelligent automation engine script is deployed together with the workflow logic to automatically process changes in user requests and send the processing results to the virtual machine monitor. The virtual machine monitor controls the resource pool to allocate or reclaim the corresponding IT resources. When cloud users increase requests, the automatic extension listener will send a signal to the intelligent automation engine to execute the script. After the script runs, the virtual machine monitor will allocate more IT resources from the resource pool to the virtual machine so that the increased workload can be processed.

This type of cloud architecture can also include the following additional mechanisms.